OnSports Business Report

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Exploratory Data Analysis

Context

Fantasy sports are online gaming platforms where participants draft and manage virtual teams composed of real professional sports players. Based on the real-world performance of these players, participants earn points on the fantasy sports platform for every match. The goal is to assemble the best possible team within a fixed budget to score the maximum fantasy points, with users competing against each other across an entire league or season. Some fantasy sports platforms involve real financial investments, offering participants the chance to win monetary rewards as well as free matchday tickets on a periodic basis.

The fantasy sports market has experienced tremendous growth in recent years, with a valuation of $18.6 billion in 2019. The football (soccer) segment led the market in 2019 with over 8 million participants worldwide and is expected to maintain its dominance in the coming years. Digitalization is one of the key factors driving the growth of the fantasy sports industry, as it enables participants to compete globally and test their strategic skills. With the rapid increase in smartphone usage and the widespread availability of fantasy sports apps, the market is projected to continue its global surge, reaching a valuation of $48.6 billion by 2027.

Objective

OnSports is a fantasy sports platform offering fantasy leagues across multiple sports and has seen a steady increase in participants globally over the past five years. Each player on the platform is assigned an

initial price, which fluctuates over time based on real-world performance. With the new English Premier League season approaching, OnSports has collected data from the previous season and seeks to analyze it to determine the starting price of each player for the upcoming season.

They have hired you as a data scientist to perform a cluster analysis aimed at identifying players of varying potential based on their previous season's performance. This analysis will help the company recognize patterns in player performances and fantasy returns, enabling them to set fair and data-driven prices for each player in the upcoming football season.

Data Dictionary

- **Player_Name**: Name of the player
- **Club**: Club the player represents
- **Position**: Position the player occupies on the field
- **Goals_Scored**: Number of goals scored by the player in the previous season
- **Assists**: Number of passes made by the player that directly led to goals in the previous season
- **Total_Points**: Total number of fantasy points earned by the player in the previous season
- **Minutes**: Total number of minutes played by the player in the previous season
- **Goals_Conceded**: Number of goals conceded by the player in the previous season
- **Creativity**: A computed score that measures a player's ability to create goal-scoring opportunities for others

- **Influence**: A computed score assessing a player's overall impact on a match, considering both direct and indirect contributions to match outcomes
- **Threat**: A computed score indicating a player's likelihood of scoring goals
- **Bonus**: Total bonus points received (awarded to the top three performing players in each match — 3 points for first, 2 for second, and 1 for third, based on a range of statistical measures)
- **Clean_Sheets**: Number of matches in which the player's team did not concede any goals in the previous season

Dataset Overview and Initial Exploration

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
Player_Name	476	476	Willy Boly	1	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Club	476	17	Arsenal	30	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Position	476	4	Midfielder	195	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Goals_Scored	476.0	NaN	NaN	NaN	1.907563	3.455562	0.0	0.0	0.5	2.0	23.0
Assists	476.0	NaN	NaN	NaN	1.752101	2.708563	0.0	0.0	0.0	2.0	14.0
Total_Points	476.0	NaN	NaN	NaN	58.516807	51.293559	0.0	10.0	48.0	94.25	244.0
Minutes	476.0	NaN	NaN	NaN	1336.909664	1073.773995	0.0	268.75	1269.5	2256.25	3420.0
Goals_Conceded	476.0	NaN	NaN	NaN	19.157563	15.946171	0.0	4.0	18.0	31.0	68.0
Creativity	476.0	NaN	NaN	NaN	195.97605	251.478541	0.0	8.3	96.95	296.95	1414.9
Influence	476.0	NaN	NaN	NaN	294.617647	267.779681	0.0	46.5	233.1	499.5	1318.2
Threat	476.0	NaN	NaN	NaN	224.962185	318.240377	0.0	5.75	104.5	298.25	1980.0
Bonus	476.0	NaN	NaN	NaN	4.718487	6.252625	0.0	0.0	2.0	7.0	40.0
Clean_Sheets	476.0	NaN	NaN	NaN	4.745798	4.394312	0.0	0.0	4.0	8.0	19.0

Figure 1: Statistical Summary

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 476 entries, 0 to 475
Data columns (total 13 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   Player_Name     476 non-null    object
 1   Club            476 non-null    object
 2   Position        476 non-null    object
 3   Goals_Scored    476 non-null    int64
 4   Assists         476 non-null    int64
 5   Total_Points    476 non-null    int64
 6   Minutes         476 non-null    int64
 7   Goals_Conceded  476 non-null    int64
 8   Creativity      476 non-null    float64
 9   Influence       476 non-null    float64
 10  Threat          476 non-null    int64
 11  Bonus           476 non-null    int64
 12  Clean_Sheets    476 non-null    int64
```

Figure 2: Data Overview

The dataset used for this project contains information on 476 football players from the English Premier League, encompassing 13 variables that capture both attacking and defensive performance metrics. These include standard statistics such as Goals Scored, Assists, Minutes Played, and Clean Sheets, as well as composite fantasy-performance indicators like Creativity, Influence, and Threat.

A preliminary inspection confirms that the dataset is well-structured and analysis-ready. Three columns — Player Name, Club, and Position — are categorical, while the remaining ten are numeric. This balance of qualitative and quantitative data provides a solid foundation for both descriptive analysis and clustering.

The dataset contains records from 17 unique clubs and four player positions (Forward, Midfielder, Defender, Goalkeeper). This diversity ensures representation across different tactical roles and playing styles, which is critical for meaningful segmentation later.

Data Quality Checks

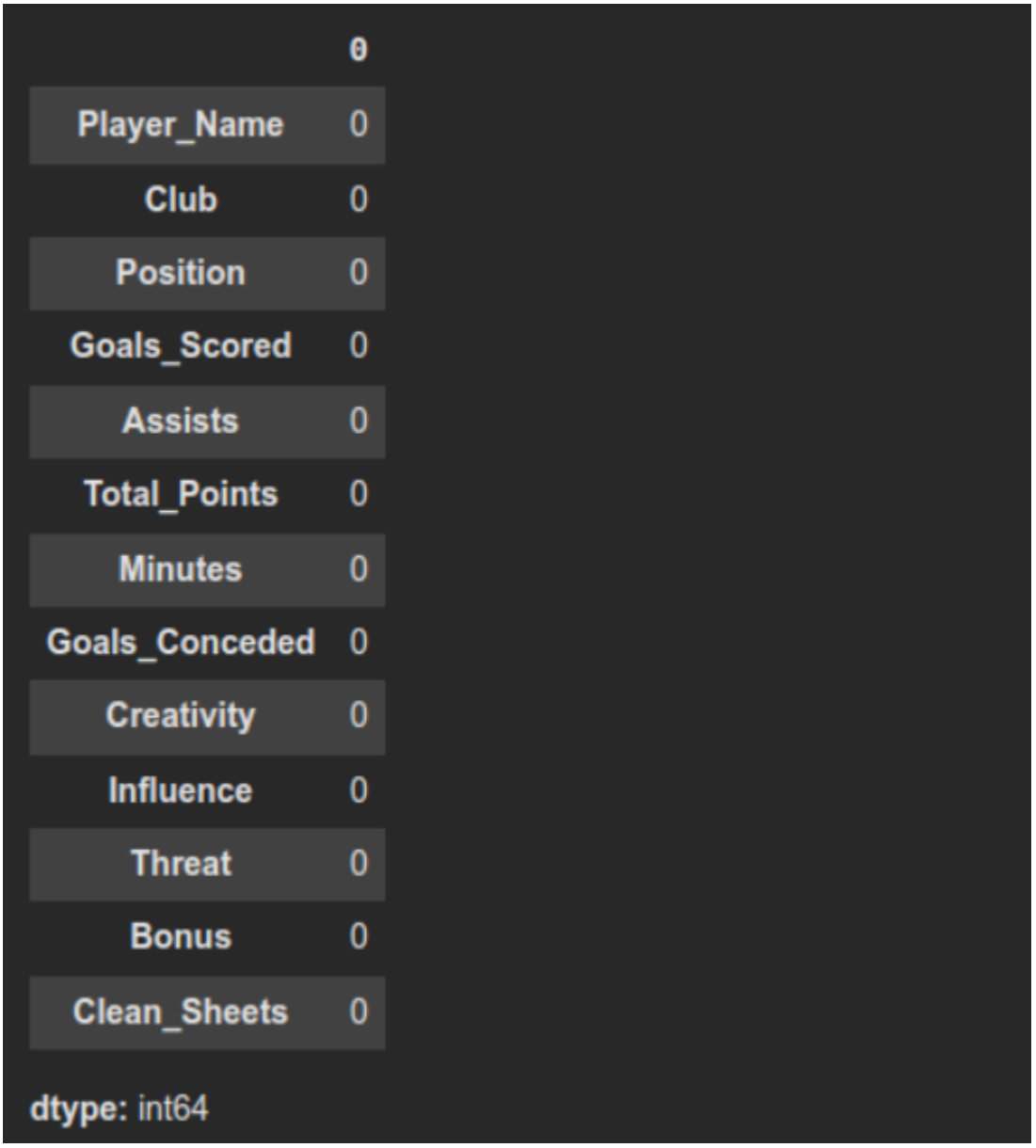


Figure 3: Duplicate values check

A series of integrity checks confirmed the reliability of the dataset:

Duplicate Records:

No duplicate entries were found, indicating that each player is represented uniquely.

Missing Values:
All columns are complete with no missing values, eliminating the need for imputation or additional data cleaning before analysis.

Data Types:
All numeric features, such as goals, assists, minutes, and composite indices like threat and creativity, are correctly stored as numerical variables. This ensures compatibility with clustering algorithms and statistical techniques.

These results collectively indicate that the dataset is of high quality and requires minimal preprocessing — a strong foundation for further feature engineering and clustering analysis.

Statistical Summary and Initial Observations

A statistical overview of the dataset reveals considerable variability in player performance:

- The average goals scored per player is around 1.9, with a maximum of 23, highlighting the gap between elite forwards and the rest of the league.
- Assists average about 1.75, peaking at 14, again showing a skewed distribution dominated by top playmakers.
- Total fantasy points average approximately 58.5 but with a large standard deviation, suggesting wide disparities in overall performance.
- Minutes played vary greatly, from none to over 3,000, indicating differences in playing time between regular starters and occasional substitutes.

- Creativity, Influence, and Threat show very high standard deviations, consistent with their nature as composite indices reflecting diverse player contributions.
- Clean Sheets range from 0 to 19, mostly reflecting defensive and goalkeeping performance.

Interpretation

From these initial diagnostics, several insights emerge:

- **Right-skewed distributions:** A small group of high-performing players dominate metrics like goals, assists, and total points, while most players cluster around lower values.
- **Position-driven variation:** Offensive and defensive attributes differ sharply across positions, suggesting positional context will be crucial for meaningful cluster interpretation.
- **Data readiness:** The dataset is already suitable for analysis and modeling, requiring no additional cleaning and ready for scaling and clustering.

Univariate Analysis

1. Goals_Scored

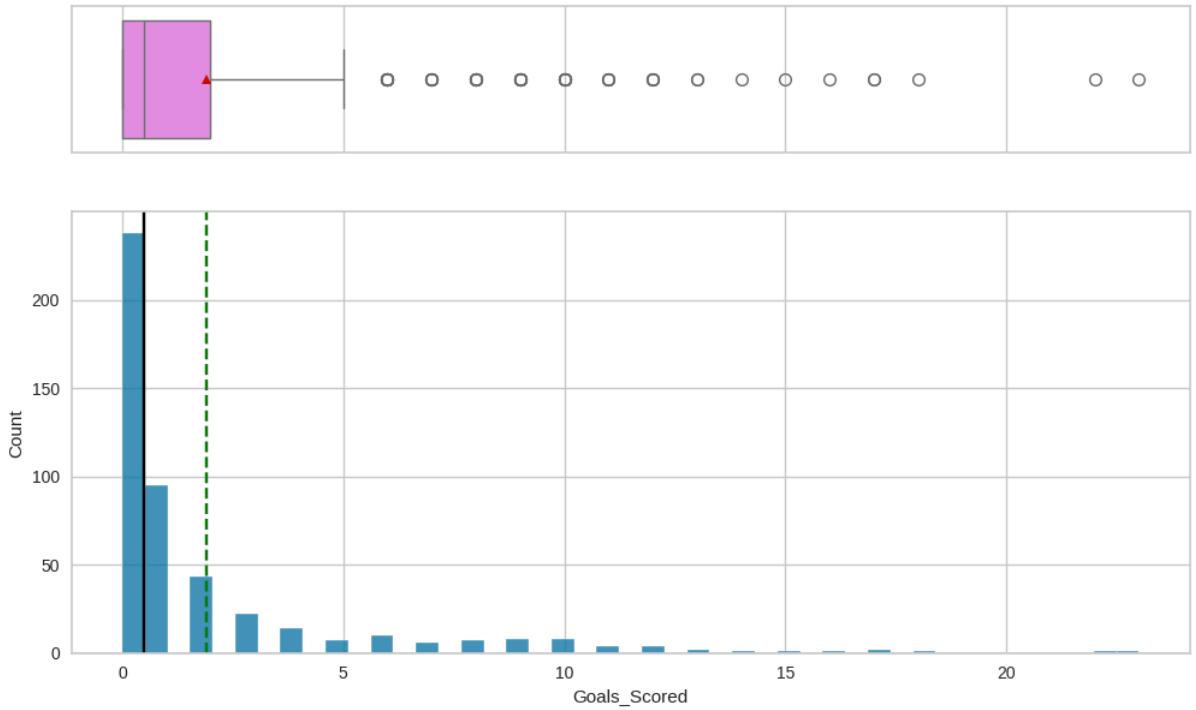


Fig.4: Barplot and histogram of Goals_Scored

The distribution of the **Goals Scored** variable reveals a highly **right-skewed pattern**, with the majority of players scoring very few goals and only a small number achieving double-digit tallies. The histogram shows that a large cluster of players scored **zero to one goal** throughout the season, reflecting the dominance of defensive players and substitutes who have limited attacking contributions.

The boxplot reinforces this observation by displaying a **tight interquartile range (IQR)** concentrated near the lower end of the scale, and numerous **outliers** extending towards the right. These outliers represent a small set of elite attackers and goal-scoring midfielders whose exceptional performance heavily influences the overall

distribution. Examples include top forwards who scored between **10 and 23 goals**, well beyond the upper whisker boundary.

This skewness is expected in football performance data, where scoring opportunities are position-dependent and heavily influenced by team tactics. Forwards and attacking midfielders naturally contribute more to goal counts, whereas defenders and goalkeepers remain close to zero.

The clear imbalance in the distribution emphasizes the importance of **normalization** before clustering, ensuring that the model does not overemphasize rare but extreme goal-scoring performances. It also suggests that **per-90-minute metrics** and composite indices such as *Threat* and *Influence* will be essential for producing a fair comparison across positions and playing times.

2. Assists

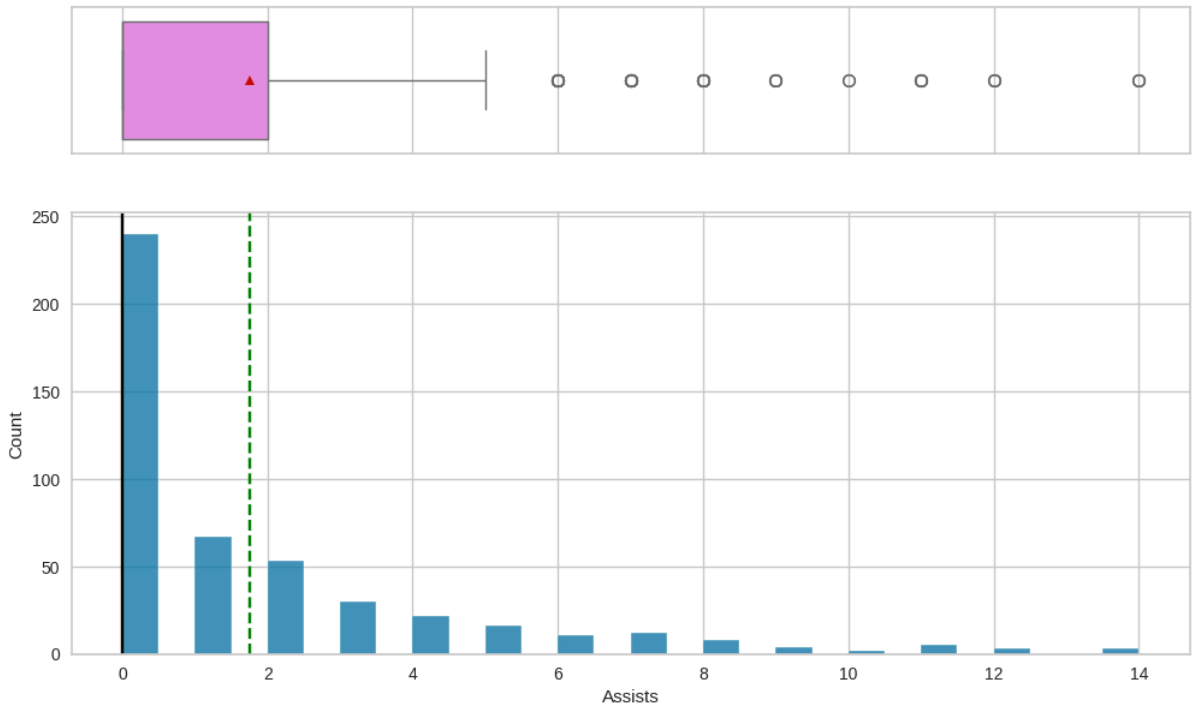


Fig.5: Histogram and Boxplot of Assists

The **Assists** variable displays a distribution that is strongly **right-skewed**, mirroring the pattern seen in *Goals Scored*. The majority of players registered **zero to one assist** across the season, while a much smaller group achieved consistently higher numbers, extending the upper tail of the distribution.

The histogram shows a sharp concentration at the lower end, with nearly half the players recording no assists at all. The boxplot reinforces this, where the median lies close to zero and a number of outliers stretch up to around **14 assists**, representing the top creative players in the league—typically attacking midfielders and wingers responsible for playmaking and delivering key passes.

The narrow interquartile range indicates that **most players contribute modestly in terms of assists**, and the variation beyond the 75th percentile is largely driven by a select group of elite creators. This pattern underscores how creative output in football is not evenly distributed but rather concentrated among specific positions and player archetypes.

From an analytical standpoint, the pronounced skew highlights the need for **feature scaling and normalization** before applying clustering algorithms. Without it, high-assist outliers could disproportionately influence the model's cluster boundaries. In practical terms, it also reinforces that assist potential is a defining feature for identifying attacking clusters within the dataset, distinguishing playmakers and forwards from defensive players who rarely record assists.

3. Goals_Conceded

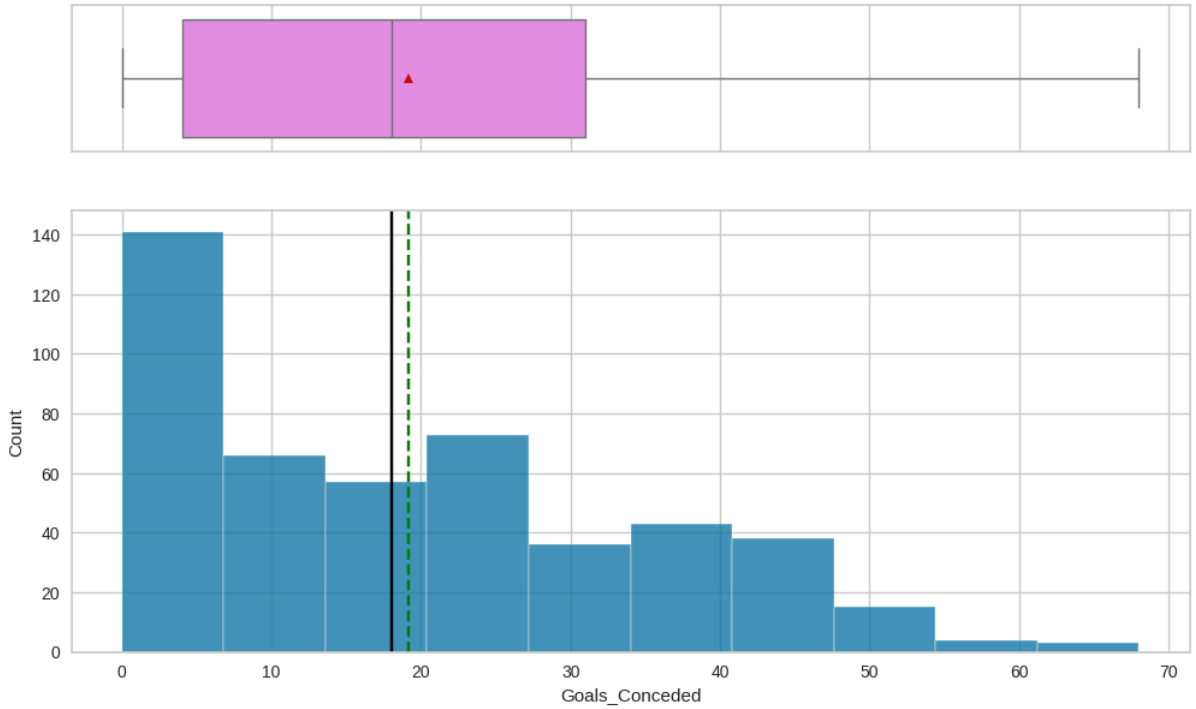


Fig.6: Histogram and Boxplot of Goals_Conceded

The Goals Conceded variable exhibits a much broader and more evenly spread distribution compared to the offensive metrics analyzed earlier. While there is still some concentration at the lower end of the scale, the histogram reveals that values are dispersed across the full range, extending up to nearly 70 goals conceded. This pattern aligns with the expectation that defenders and goalkeepers—who play full matches—will record higher totals in this category, whereas attacking players and substitutes typically concede fewer goals due to limited defensive involvement.

The median value sits close to 18 goals conceded, and the boxplot demonstrates a relatively wide interquartile range, indicating substantial variability across players. Unlike goal or assist distributions, which were heavily right-skewed, Goals Conceded appears more balanced, albeit

with a gentle right tail reflecting defenders in struggling teams who endured frequent defensive lapses.

A key insight from this plot is the positional dependency of the variable. While high values generally correspond to defenders and goalkeepers, a significant number of players also show zero or near-zero goals conceded—these are primarily forwards and attacking midfielders with minimal defensive exposure. This positional distinction reinforces the need to interpret this feature contextually within clustering, ensuring that offensive and defensive players are not compared directly on this raw metric.

From a modeling perspective, the variability and positional contrast make Goals Conceded an informative feature for distinguishing defensive roles and performance levels. However, before clustering, it will be scaled and potentially normalized to prevent its magnitude from overpowering more subtle offensive attributes such as assists or creativity.

4. Clean_Sheets

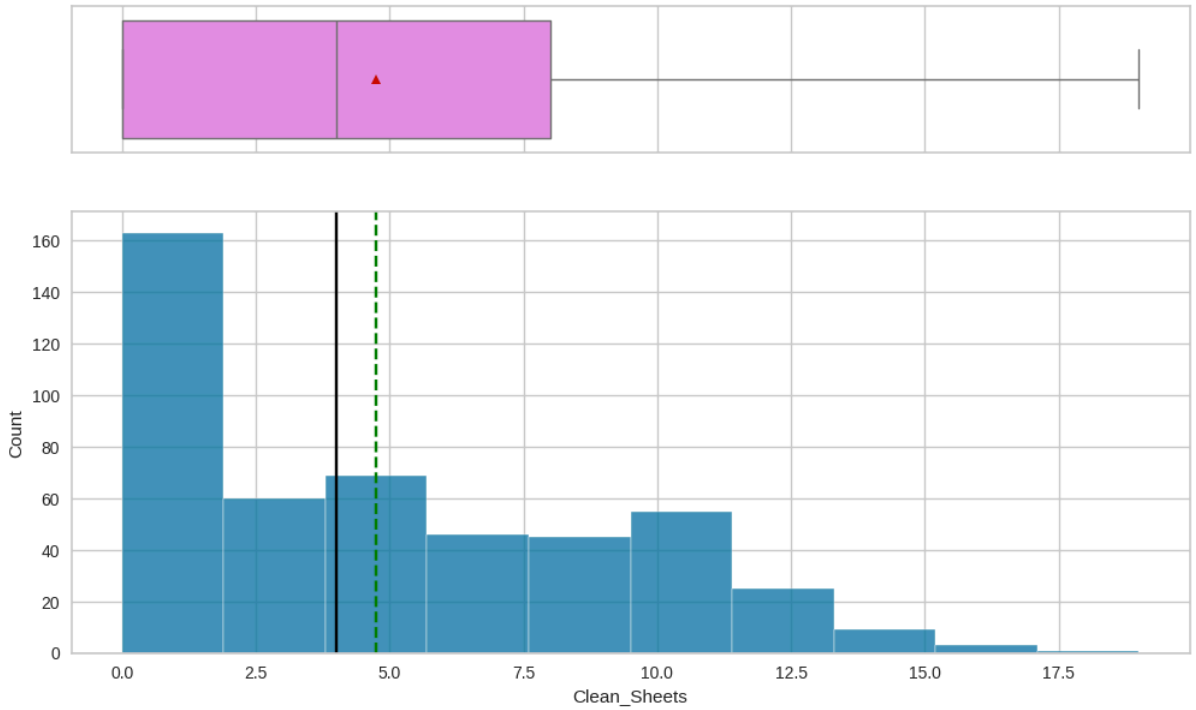


Fig.7: Histogram and Boxplot of Clean_Sheets

The **Clean Sheets** variable demonstrates a moderately right-skewed distribution, with most players achieving few or no clean sheets during the season. The histogram shows a clear concentration near zero, gradually tapering off as the number of clean sheets increases. The boxplot further confirms this, displaying a relatively wide spread with several players attaining double-digit clean sheets — primarily goalkeepers and defenders from high-performing teams with strong defensive records.

The median number of clean sheets lies around **four**, while the upper quartile extends toward **eight or nine**, indicating that only a limited subset of players consistently maintain clean sheets across multiple matches. The long right tail represents standout defensive performers,

some of whom recorded **15–19 clean sheets**, reflecting exceptional team defense and consistent starting minutes.

This distribution aligns closely with the positional dynamics of football: attackers and midfielders naturally record fewer clean sheets, while defenders and goalkeepers dominate this metric. Consequently, *Clean Sheets* serves as a strong differentiator for defensive players in the clustering process. It helps capture the contribution of players whose fantasy value is driven less by scoring and more by defensive solidity.

From a modeling standpoint, the variability across positions necessitates **scaling and normalization** before clustering, to ensure that this defensive metric holds proportional influence relative to offensive attributes like goals and assists. Nevertheless, the clean sheet count remains an essential feature for identifying high-value defensive clusters in the dataset.

5. Minutes

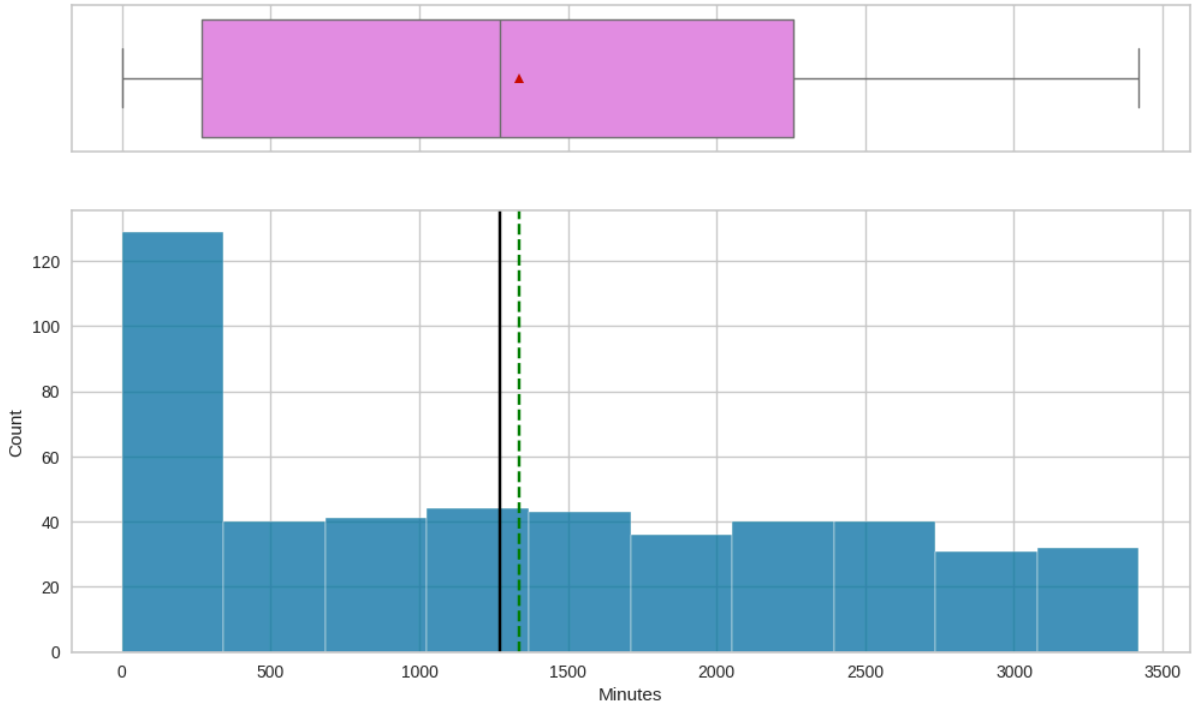


Fig.8: Histogram and Boxplot of Minutes

The **Minutes Played** variable exhibits a wide and somewhat uniform distribution with a noticeable spike at the lower end. A significant portion of players recorded **less than 500 minutes** during the season, reflecting substitutes, reserve players, or individuals who featured intermittently due to injuries or tactical rotations. In contrast, another large group accumulated well over **2,500 minutes**, representing consistent starters who played nearly full matches across the season.

The boxplot reveals a **broad interquartile range**, indicating high variability in playing time across the player pool. The median lies around **1,300 minutes**, suggesting that half of the players featured for roughly half of the available match minutes. The upper quartile extends close to **2,200 minutes**, while a few outliers approach the maximum of **3,420 minutes**, typical for players who appeared in nearly every game.

The histogram's shape illustrates two key insights. First, the **bimodal nature**—with concentrations at both low and high minute counts—reflects the natural divide between core starters and rotational or backup players. Second, this uneven participation significantly influences cumulative metrics such as goals, assists, and total points, emphasizing the need for normalization when comparing player performance.

From a modeling perspective, this variable is critical for interpreting player contribution levels. However, since raw minutes can bias performance-based clustering (favouring those who simply played more), they will be transformed into **per-90-minute metrics** (e.g., goals per 90, assists per 90, and points per 90). This adjustment ensures that clustering reflects **performance efficiency** rather than mere playing time, allowing fair comparisons across all player types.

6. Total_Points

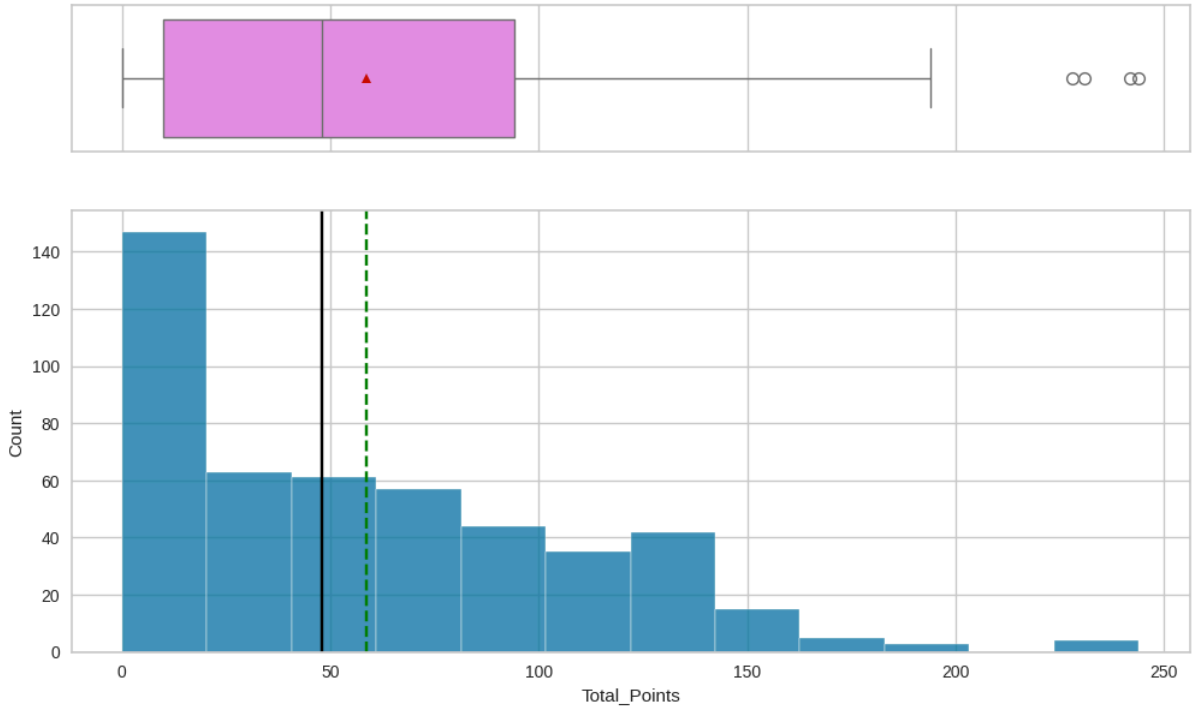


Fig.9: Histogram and Boxplot of Total_Points

The **Total Points** variable displays a distinct **right-skewed distribution**, characteristic of fantasy football scoring systems where a small number of elite players dominate the leaderboard. The histogram shows that most players accumulated fewer than **70 total points** across the season, while a limited group achieved totals exceeding **150 points**, and only a few exceptional performers crossed the **200-point** threshold.

The boxplot illustrates this disparity vividly — the interquartile range is concentrated in the lower half of the scale, while several **outliers** extend far beyond the upper whisker, representing high-impact players who consistently contributed through goals, assists, clean sheets, and bonus points. These outliers typically correspond to regular starters from top-tier clubs or key attacking players with heavy involvement in match outcomes.

The median total points hover around **48–50**, aligning with rotational or mid-tier players who contributed occasionally but not consistently. The wide range and large standard deviation underscore the heterogeneous nature of the player pool, spanning from peripheral squad members to star performers.

From an analytical standpoint, *Total Points* serves as an excellent **summary indicator of overall fantasy value**. However, since it aggregates multiple underlying factors (offensive, defensive, and disciplinary), it will later be complemented by decomposed performance metrics such as *Goals Scored*, *Assists*, *Clean Sheets*, and *Bonus Points* to provide more granular clustering insights.

Moreover, because *Total Points* is influenced by the number of minutes played, **normalizing it on a per-90-minute basis** ensures fair comparisons between regular starters and players with limited appearances. This normalization step will allow the clustering model to group players based on **performance efficiency** rather than total match exposure — a crucial distinction for fantasy pricing and talent categorization.

7. Creativity

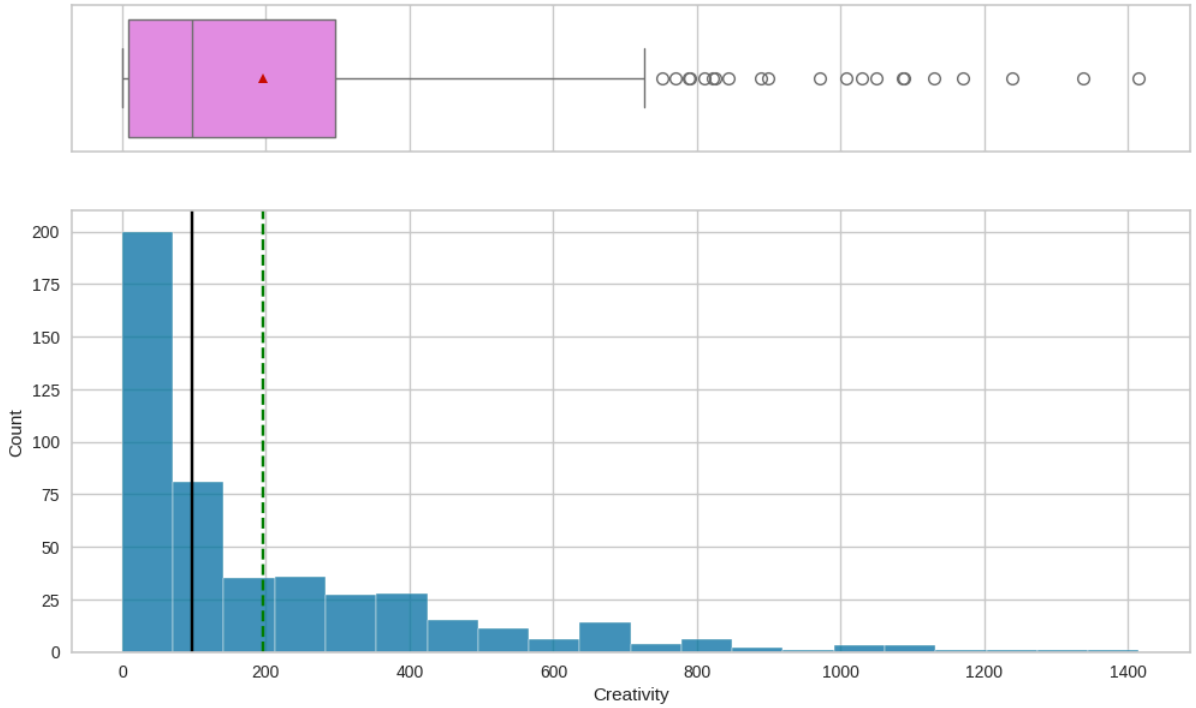


Fig.10: Histogram and Boxplot of Creativity

The **Creativity** variable exhibits a highly **right-skewed distribution**, indicating that a small subset of players dominate in generating goal-scoring opportunities. The histogram shows that the majority of players possess *Creativity scores* clustered near the lower end of the scale, typically below **150**, while a select group extends well beyond **500**, with the most creative players reaching scores above **1,000**.

The boxplot supports this observation, displaying a dense concentration near the lower quartile and numerous **outliers** on the upper end. These outliers represent playmakers—often attacking midfielders and advanced fullbacks—who are heavily involved in their team's offensive buildup through key passes, crosses, and assists.

The median Creativity value lies well below the mean, reinforcing the presence of extreme high-value players skewing the distribution. The

interquartile range is relatively narrow, suggesting that while most players contribute minimally in creative play, the few at the top have an outsized impact on team offense.

From a business and fantasy sports perspective, *Creativity* is a crucial differentiator in identifying **assist-driven value players**—those whose contributions may not directly translate into goals but still yield consistent fantasy points through their influence on attacking sequences.

In the context of clustering, the skewed distribution highlights the need for **feature scaling** to prevent high-creative players from dominating cluster formation. After normalization, this variable will help distinguish **playmakers and attacking orchestrators** from more defensively oriented or goal-focused players, providing an important layer of nuance in player segmentation and pricing strategy.

8. Influence

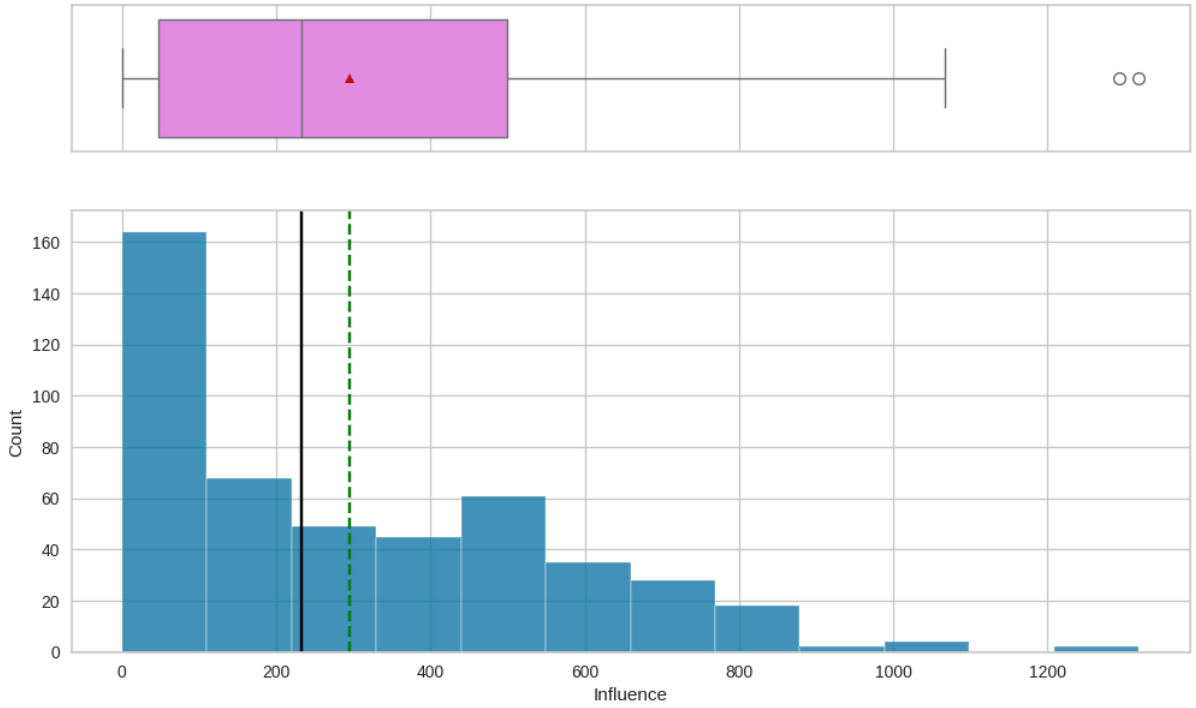


Fig.11: Histogram and Boxplot of Influence

The **Influence** variable presents a moderately right-skewed distribution, indicating that while most players exert limited overall match impact, a smaller group stands out with significantly higher influence scores. The histogram shows that the majority of players cluster below **300**, while a gradual tapering occurs toward the upper end, with a few exceptional individuals exceeding **1,000**. These high values correspond to players who are pivotal to their teams — often captains, central playmakers, or dominant defenders whose contributions shape game outcomes.

The boxplot mirrors this trend, with a long right tail and two clear outliers representing the most impactful players in the league. The median value lies well below the mean, confirming that a small number of high-impact performers elevate the overall average. This disparity highlights the uneven distribution of influence in football, where leadership,

decision-making, and consistency are concentrated among a handful of elite players.

Conceptually, *Influence* is a composite measure that integrates actions both direct (such as goals and assists) and indirect (such as defensive interventions, key passes, or tactical importance). Its significance transcends position — goalkeepers, defenders, midfielders, and forwards can all exhibit high influence depending on their role within team dynamics.

From a modeling perspective, *Influence* acts as a **holistic measure of player importance**, blending both offensive and defensive contributions. Its right-skewed distribution suggests that normalization will be essential to prevent outlier domination during clustering. Post-scaling, *Influence* will play a central role in distinguishing **high-impact core players** — the backbone of their teams — from **supporting or rotational players** with lower overall game involvement.

9. Threat

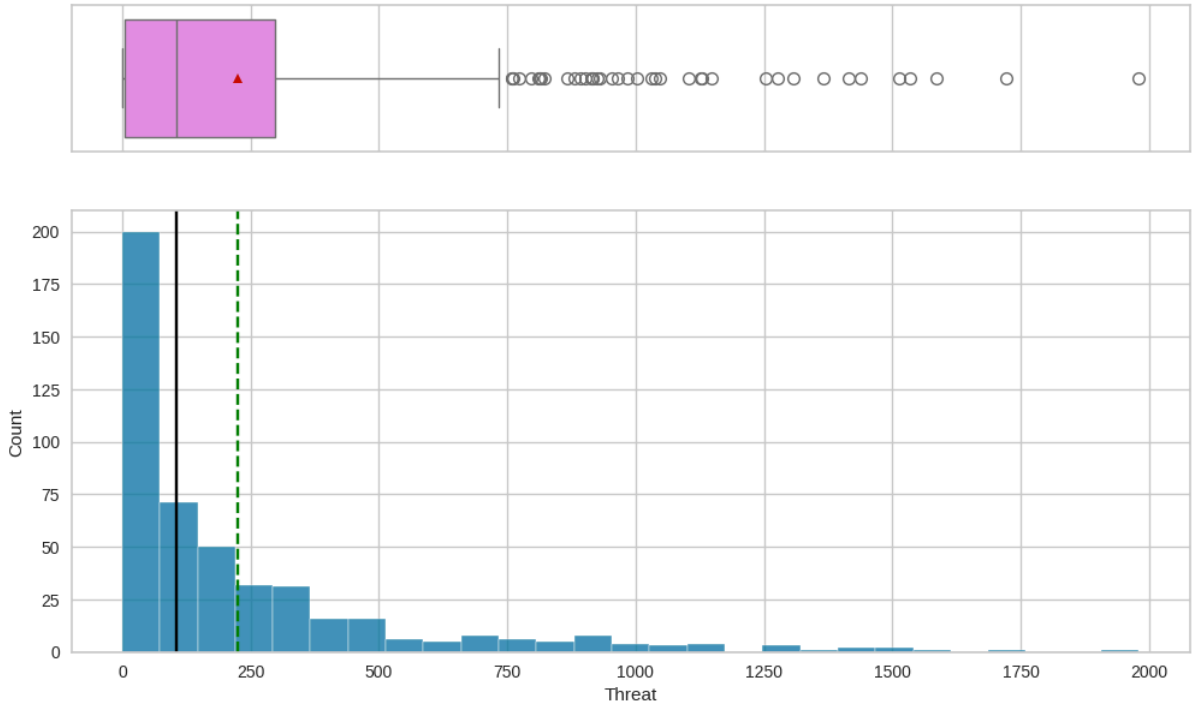


Fig.12: Histogram & boxplot for Threat

The **Threat** variable exhibits a distinctly **right-skewed** distribution, reflecting the concentrated nature of goal-scoring opportunities in football. The histogram shows that a large portion of players have *Threat scores* below **200**, while a small subset of elite attackers achieve extremely high values, extending well beyond **1,000** and even approaching **2,000** in rare cases. These high-end performers represent the league's most aggressive offensive players—typically forwards and advanced wingers who consistently register shots, chances, and expected goals.

The boxplot further emphasizes this disparity, with a dense cluster of players near the lower quartile and a substantial number of **outliers** stretching far to the right. The median lies close to the bottom of the scale, confirming that the majority of players contribute little in terms of

direct goal-scoring threat, while only a minority dominate in offensive metrics.

This imbalance is expected, as the *Threat* metric captures players' involvement in attacking phases—shot attempts, positioning in the final third, and proximity to scoring situations. Forwards and attacking midfielders naturally accumulate higher scores, while defenders and goalkeepers remain near zero, contributing minimally to attacking play.

From an analytical standpoint, *Threat* serves as one of the most **discriminative variables** in the dataset, clearly distinguishing between attack-oriented and defense-oriented player archetypes. However, its high variance and extreme skew require **scaling and normalization** before clustering to ensure balanced feature influence.

In a business context, this metric has strong implications for **fantasy player pricing**. High-threat players tend to be premium assets, as their consistent involvement in scoring chances translates directly to higher point potential. When integrated into clustering, *Threat* will help identify **high-risk, high-reward attackers** versus **steady, lower-risk defensive or midfield options**, enabling more nuanced segmentation for player valuation and strategy optimization.

10. Bonus

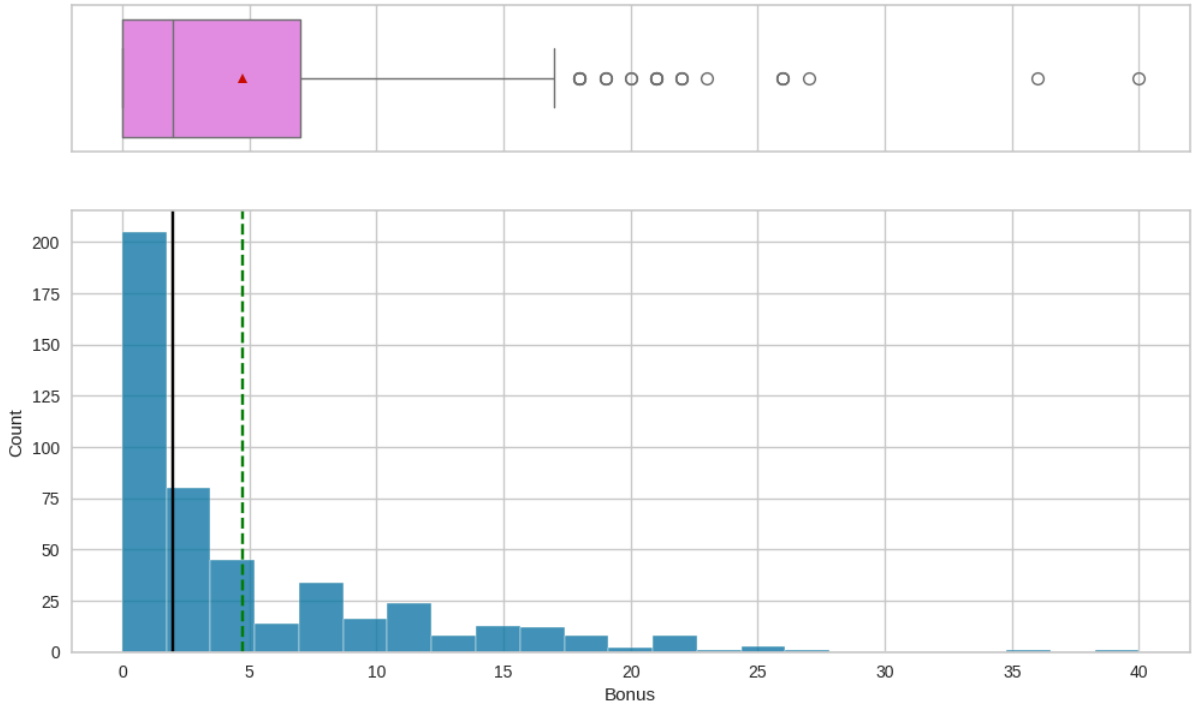


Fig.13: Histogram & boxplot for Bonus

The **Bonus Points** variable shows a markedly **right-skewed distribution**, illustrating that most players receive very few or no bonus points over the course of a season, while a limited group of consistently high-performing players earn substantially more. The histogram reveals that the majority of players lie below **5 bonus points**, with the frequency dropping sharply beyond that range. A small but notable number of players reach values above **20**, and the highest performers accumulate **up to 40 bonus points**, reflecting exceptional match impact and consistency.

The boxplot highlights this inequality clearly — a dense concentration of observations near zero, followed by numerous **outliers** extending to the right. These outliers represent players who repeatedly ranked among the

top three performers in individual matches, earning bonus points due to sustained excellence in both offensive and defensive contributions.

Bonus points in fantasy scoring systems are often tied to a player's **overall influence and efficiency** — factors such as goals, assists, clean sheets, and match-winning contributions. Therefore, while this metric correlates strongly with total points, it also provides additional insight into **match-to-match consistency** and the ability to deliver peak performances under pressure.

From an analytical perspective, the skewed nature of *Bonus Points* once again underscores the need for **scaling prior to clustering** to ensure balance among features. Within the broader modeling framework, this variable serves as a valuable indicator for identifying **elite performers** whose fantasy value extends beyond raw statistical contributions — players who are not just productive, but also **decisive and consistent**.

In business terms, high *Bonus Point* earners represent the **premium fantasy assets**— dependable, impactful players who command higher pricing and attract greater demand among users. Conversely, the majority of low scorers reflect rotational or situational players who offer limited fantasy return.

11. Club

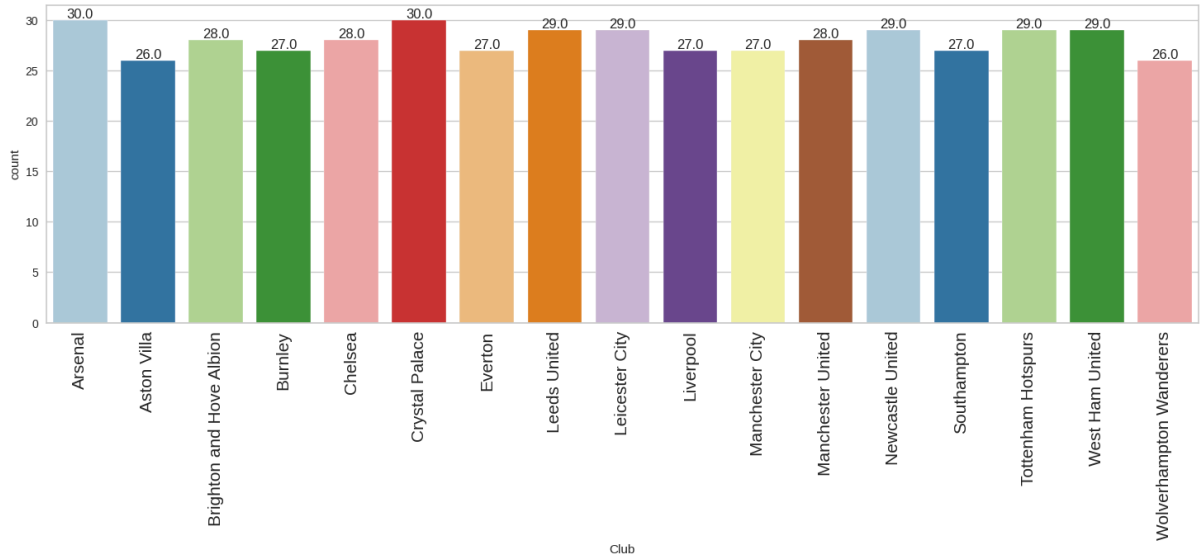


Fig.14: Barplot of Club

The **Club** variable provides a categorical overview of how players are distributed among the 17 Premier League teams in the dataset. The bar chart shows a **remarkably balanced representation** across all clubs, with each contributing roughly between **26 and 30 players**.

This even spread mirrors the typical squad composition of Premier League teams, where clubs generally maintain squads of 25 to 30 registered players per season. Both **Arsenal** and **Crystal Palace** have the highest representation with **30 players each**, while **Aston Villa** and **Wolverhampton Wanderers** have the fewest, at **26 players**.

Such balance in representation ensures that the dataset provides a **comprehensive and unbiased overview of player performance** across the league. No single team dominates the dataset, which helps maintain analytical neutrality when identifying player performance patterns. It also implies that any clusters derived later will capture **player-level differences** rather than team-level biases.

From an interpretive standpoint, this uniform club distribution forms a strong foundation for subsequent analysis. It guarantees that high-performing players from smaller clubs receive equal consideration as those from top-tier teams. For OnSports, this creates a **fair and representative baseline** for assessing player value, ensuring that fantasy pricing models are not skewed toward popular or high-budget clubs.

12. Position

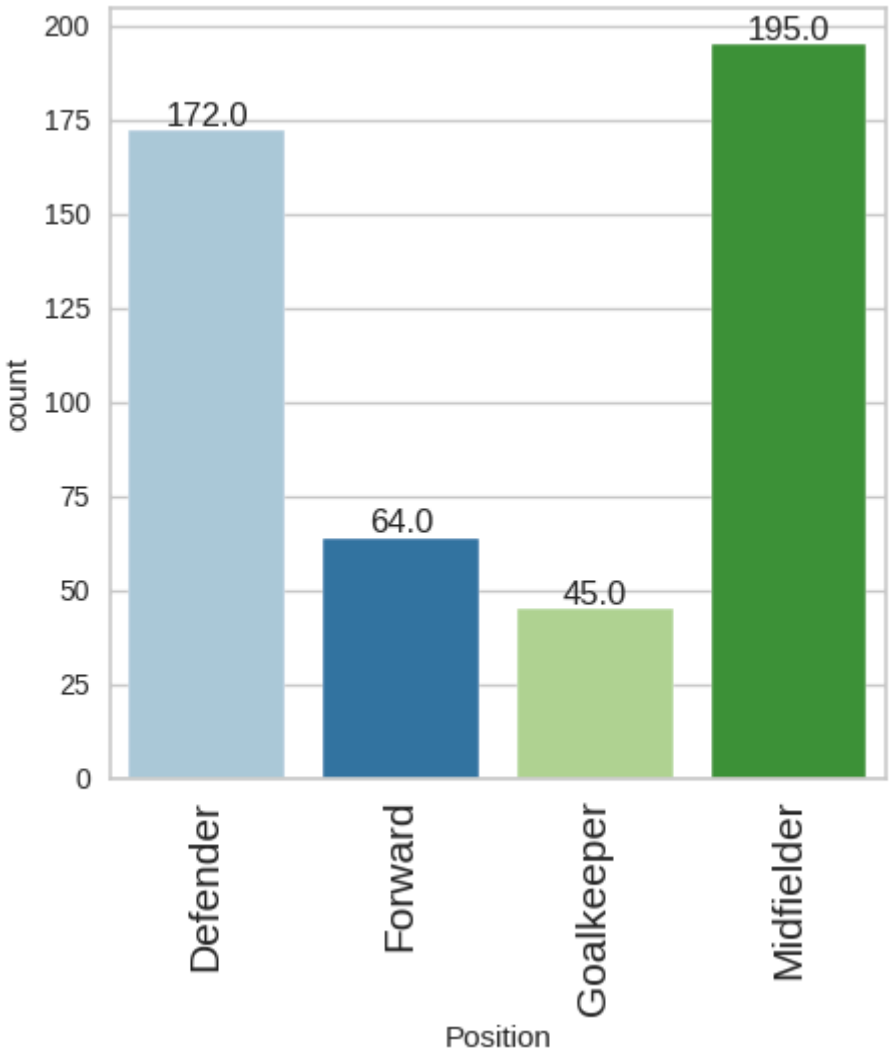


Fig.15: Barplot of Position

The **Position** variable provides an overview of how players are distributed across different roles on the pitch — namely **Goalkeeper**, **Defender**, **Midfielder**, and **Forward**. The distribution shows that **Midfielders** form the largest group, with **195 players**, followed by **Defenders** at **172 players**. **Forwards** and **Goalkeepers** make up smaller portions of the dataset, with **64** and **45 players**, respectively.

This distribution accurately reflects the **positional structure of a typical football league**, where midfielders and defenders outnumber attackers and goalkeepers due to tactical diversity and rotation depth. Midfielders often serve multiple functions — creative playmaking, defensive support, or box-to-box movement — making them the most numerous group. Defenders, too, feature heavily as teams generally field four per match, with additional substitutes and backups across multiple positions (center-back, full-back, wing-back).

In contrast, forwards and goalkeepers represent **specialized roles**. Forwards are fewer in number since clubs often rely on one or two strikers in their tactical setup, while only a handful of goalkeepers are registered per club, given the low substitution and rotation frequency in that position.

From an analytical perspective, this positional spread is beneficial for **cluster formation**, as it ensures sufficient representation of every player type. However, it also highlights the need for **contextual clustering** — the model must account for the fact that metrics like *Goals Scored* and *Clean Sheets* vary inherently by position.

For the business application, this balance ensures that **fantasy player pricing** can be designed with nuanced positional understanding —

recognizing that midfielders dominate the dataset not by bias, but by their natural prevalence and contribution diversity within team structures.

Bivariate Analysis

Heat Map for all numerical variables

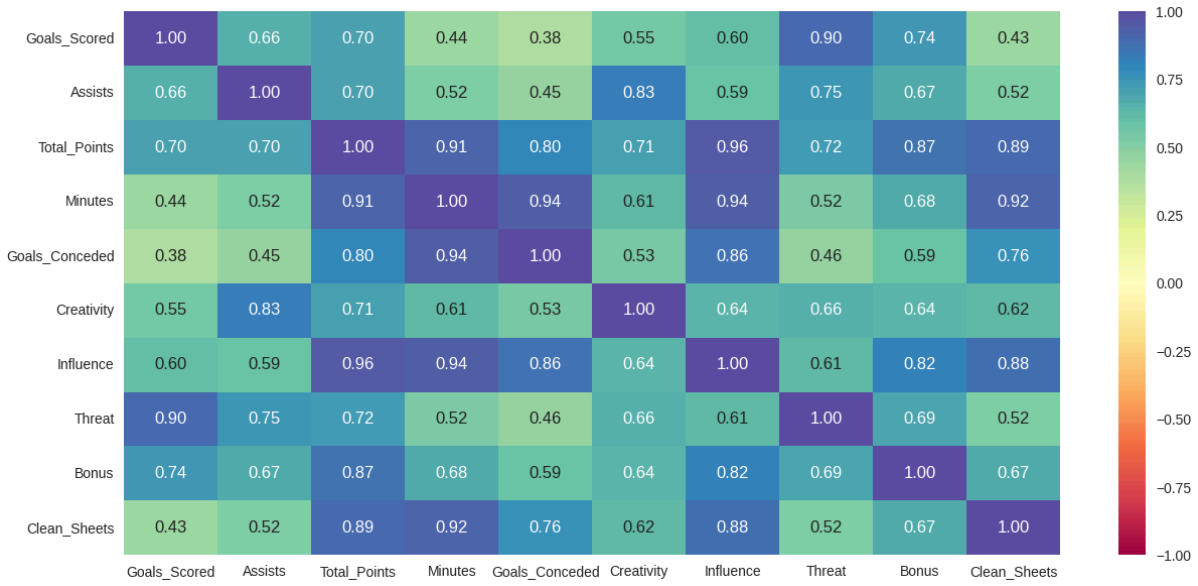


Fig.16: Heat Map for all numerical variables

The correlation heatmap provides an in-depth view of how numerical performance metrics in the dataset relate to each other. The relationships uncovered here help identify which features capture overlapping information and which offer unique insights, guiding both feature selection and interpretation for clustering.

Key Observations

1. **Strong Positive Correlations with Total Points:**
    - *Total Points* shows very high correlations with several features — most notably with *Minutes* (0.91), *Influence* (0.96), and *Clean Sheets* (0.89).
    - This pattern suggests that players who consistently feature in matches and exert significant influence tend to accumulate higher fantasy points. These attributes are thus central to

defining player value.

2. **Goals and Threat Relationship:**
    - *Goals Scored* and *Threat* have an exceptionally high correlation (0.90), confirming that the *Threat* metric effectively captures a player's goal-scoring potential.
    - Both variables are also strongly tied to *Bonus Points* (0.74) and *Total Points* (0.70), underscoring how offensive productivity directly drives fantasy performance.
3. **Assists and Creativity Link:**
    - The correlation between *Assists* and *Creativity* (0.83) is among the strongest in the dataset, which validates the *Creativity* metric's design. It accurately reflects a player's ability to generate goal-scoring opportunities that convert into assists.
    - Together, these two attributes define **playmaking strength**, which will likely form a distinct cluster representing creative midfielders and attacking fullbacks.
4. **Influence as a Central Metric:**
    - *Influence* exhibits high correlations with nearly every other variable (especially *Total Points* = 0.96 and *Goals Conceded* = 0.86).
    - This demonstrates that *Influence* serves as a composite indicator of overall player impact, blending offensive and defensive contributions — crucial for identifying all-round performers.

5. **Defensive Indicators:**
    - *Clean Sheets* correlates strongly with *Minutes* (0.92) and *Goals Conceded* (0.76), indicating that regular starters in defensive roles are more likely to accumulate clean sheets.
    - This reinforces the idea that defensive stability is a function of both playing time and team performance.
6. **Interdependence and Redundancy:**
    - Several features, such as *Influence*, *Minutes*, and *Total Points*, are so strongly correlated that including all of them without scaling or dimensionality reduction may introduce redundancy.
    - Techniques such as **Principal Component Analysis (PCA)** or selective feature weighting could later be used to balance their impact in clustering.

Interpretation

Overall, the correlation analysis highlights three major performance dimensions emerging naturally in the dataset:

- **Attacking Performance:** Defined by *Goals Scored*, *Threat*, *Bonus Points*, and *Assists*.
- **Creative Influence:** Driven by *Creativity*, *Assists*, and *Influence*.
- **Defensive Solidity:** Characterized by *Clean Sheets*, *Goals Conceded*, and *Minutes Played*.

These dimensions will form the analytical foundation for segmentation in subsequent clustering. By understanding how these metrics interrelate, the model can group players not just by their raw output, but by **holistic performance patterns** — scorers, creators, defenders, and balanced contributors.

Which team has scored the most fantasy points on average?

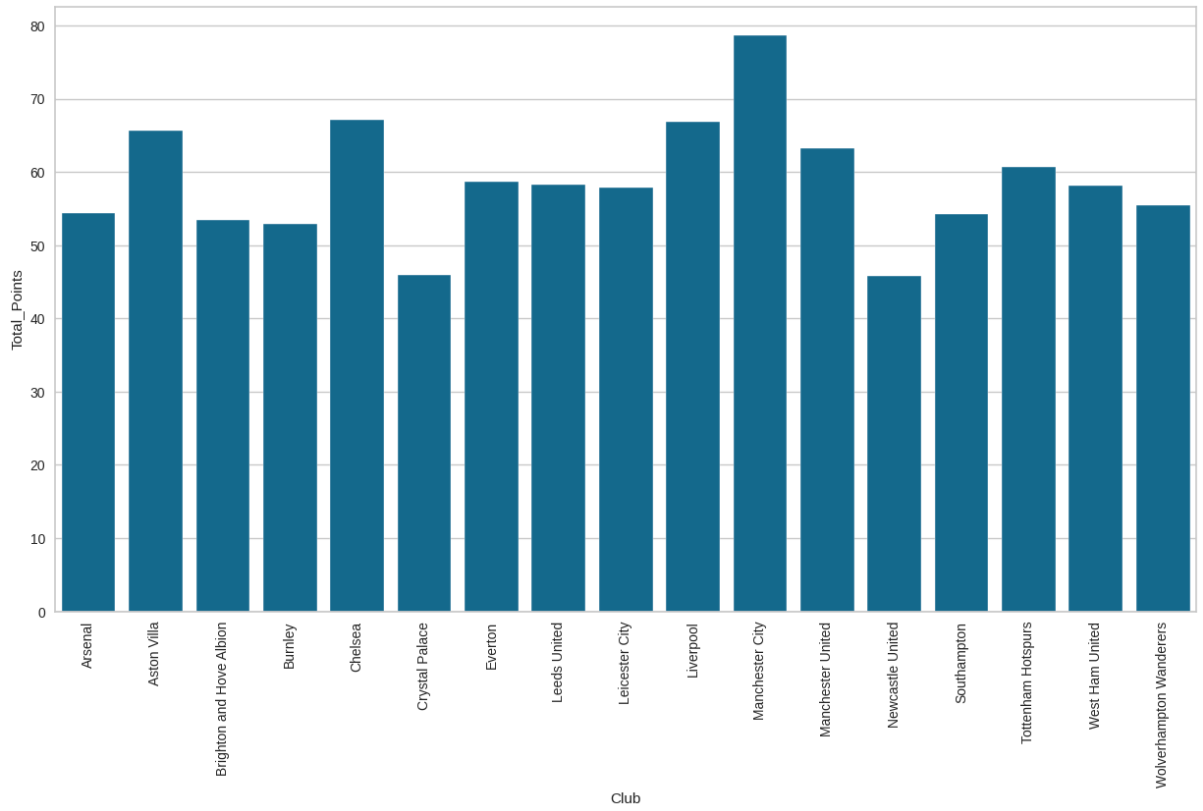


Fig.17: Club × Total_Points

The bar chart compares the **average Total Fantasy Points** earned by players across different Premier League clubs. This visualization highlights how player performance — and by extension, fantasy value — varies from team to team.

The results show a clear differentiation between clubs:

- **Manchester City** leads with the **highest average total fantasy points**, indicating that its players consistently deliver strong performances across the season. This dominance aligns with the team's real-world success, where attacking intensity and defensive stability yield higher fantasy returns.

- Close behind are **Chelsea**, **Liverpool**, and **Leicester City**, each maintaining a high average around the 65–70 point range, reflecting balanced squad contributions and strong overall team play.
- On the lower end, **Crystal Palace**, **Burnley**, and **Newcastle United** record the **lowest average total points**, typically around the mid-40s. This suggests that players from these clubs provide fewer fantasy returns on average, possibly due to limited attacking output or weaker overall team performance.

The relatively tight clustering of mid-table clubs between **55 and 65 points** shows competitive parity within the league's middle segment, where individual player contributions are less polarized.

From a business and strategic perspective, this analysis is valuable for **player pricing and user guidance** within fantasy platforms like OnSports. Players from top-performing teams like Manchester City and Liverpool should command **premium fantasy valuations**, given their consistent point generation, while those from lower-performing clubs might be priced more affordably to maintain competitive balance and user engagement.

Players in which positions tend to score more fantasy points on average?

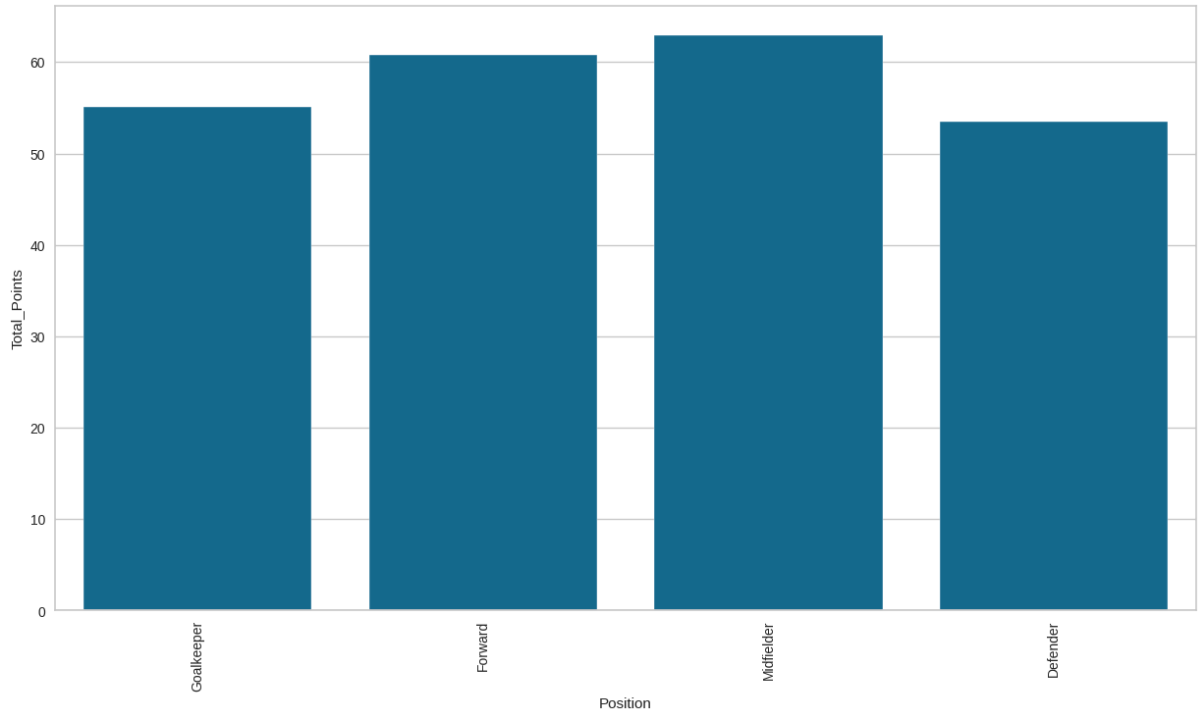


Fig.18: Position × Total_Points

The bar chart illustrates the **average total fantasy points** earned by players across the four main football positions — **Goalkeepers, Defenders, Midfielders,** and **Forwards**. This visualization helps uncover how positional roles influence overall fantasy performance.

The results indicate a clear hierarchy in average point contributions:

- **Midfielders** emerge as the top-performing group, averaging the **highest total fantasy points**. This dominance reflects their dual role in both attacking and defensive phases — contributing goals, assists, and bonus points, while also benefiting from occasional clean sheets.

- **Forwards** rank closely behind midfielders, maintaining strong averages primarily through goals, shots on target, and attacking threat metrics. Their reliance on direct scoring actions makes them highly valuable fantasy assets, though their output tends to fluctuate based on form and fixture difficulty.
- **Goalkeepers** and **Defenders** register **lower average fantasy points** compared to attacking positions. Their contributions come mainly from clean sheets, saves, and defensive bonuses, which, while consistent, are less frequent and less variable than attacking returns.
- Among defensive players, **Goalkeepers** slightly outperform **Defenders**, owing to the inclusion of save points and penalty-stop bonuses, which add steady value even in matches without clean sheets.

This pattern aligns with the structure of fantasy scoring systems, where offensive contributions (goals and assists) yield higher points per event than defensive actions. As a result, **attacking positions dominate fantasy value**, while defensive ones provide more **stability and predictability**.

From a strategic and business perspective, this insight is crucial for **fantasy player pricing and segmentation**. Midfielders and forwards should occupy **higher price tiers**, reflecting their greater potential for explosive returns, while defenders and goalkeepers should be **moderately priced** to maintain balance and encourage tactical diversity among fantasy users.

Total number of minutes played, on average, across different positions.

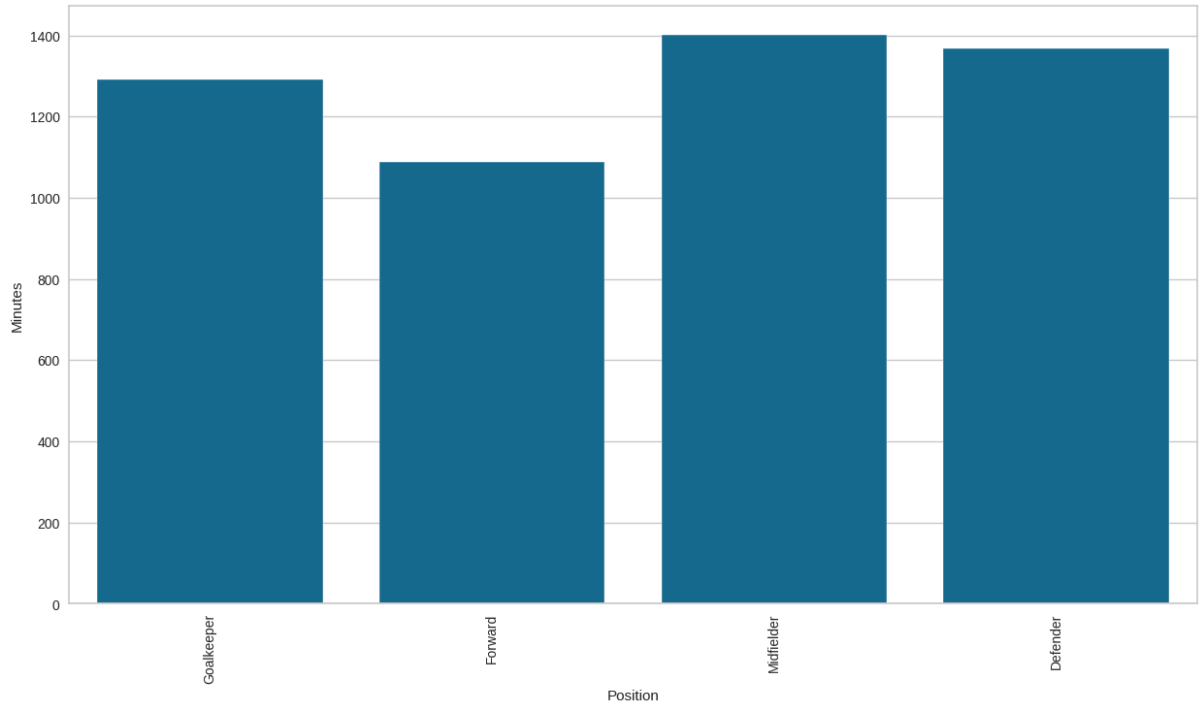


Fig.19: Position × Minutes

The bar chart presents the **average number of minutes played** by footballers in each position — **Goalkeeper**, **Defender**, **Midfielder**, and **Forward** — offering insight into playing time patterns and positional consistency throughout the season.

The findings reveal a distinct pattern of playing time distribution:

- **Midfielders** record the **highest average minutes**, around 1,400 per season. This reflects their central tactical importance — they are frequently rotated less than forwards and often participate in both offensive and defensive phases, making them indispensable across match situations.
- **Defenders** follow closely behind, also averaging well above 1,300 minutes. This consistency stems from the stability of defensive

lineups; managers prefer maintaining a fixed backline to preserve team chemistry and structural integrity.

- **Goalkeepers** show slightly fewer average minutes than outfield defenders but still maintain a **very high level of match involvement**. Since each club typically relies on a single starting goalkeeper for most matches, the high average aligns with the specialized, consistent nature of the position.
- **Forwards**, on the other hand, exhibit the **lowest average playing time** (approximately 1,080 minutes). This reflects higher rotation rates, tactical substitutions, and injury susceptibility among attacking players — as managers often vary their attacking combinations to adapt to opponents and maintain player freshness.

These results highlight that **defensive and midfield roles offer greater match-time stability**, while attacking roles, though potentially more explosive in point generation, carry higher volatility due to rotation and form dependency.

From a **fantasy football business perspective**, this insight is valuable for optimizing **price-to-performance balance**.

- Midfielders and defenders should be **priced for their reliability**, given their consistently high playing time and steady accumulation of points.
- Forwards, while capable of high peak performances, should be valued for their **impact potential rather than consistency**.
- Goalkeepers, with their dependability, offer **budget stability** and should serve as foundational picks for users seeking long-term returns.

Which team's players have secured the most bonus points, on average, last season?

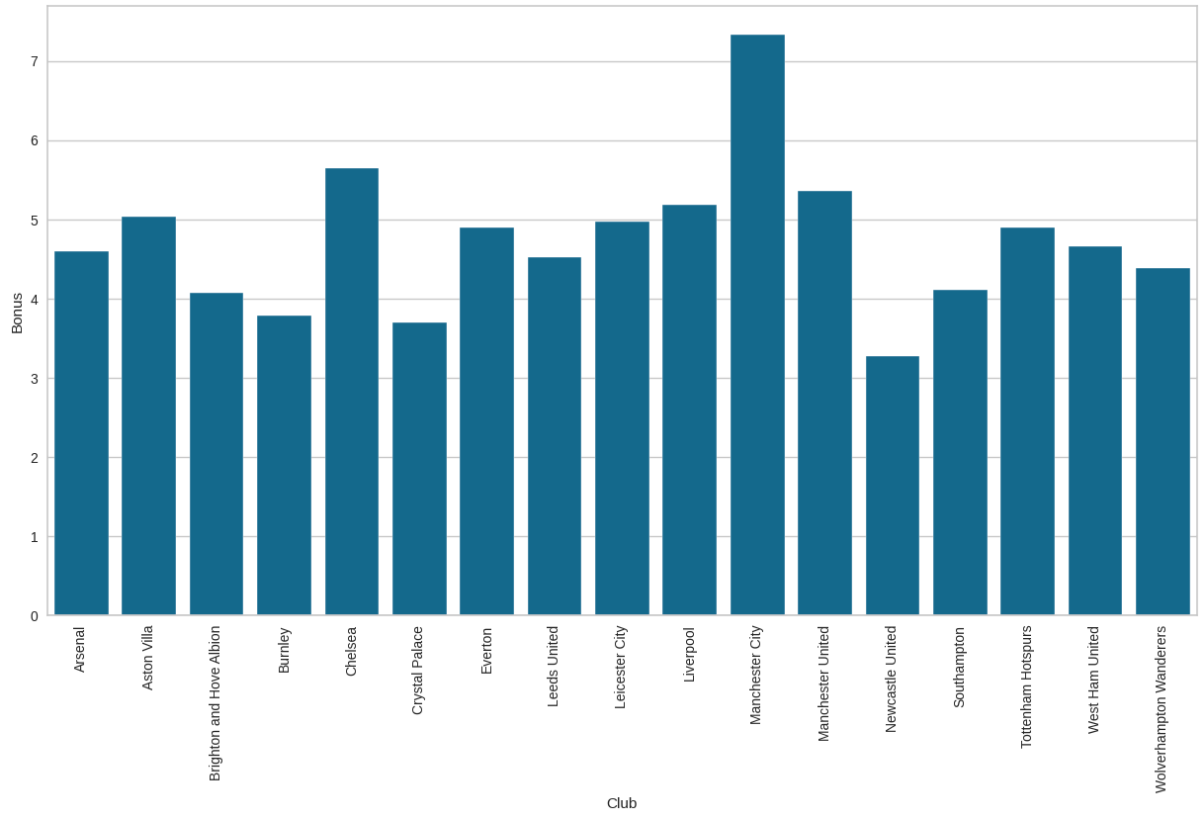


Fig.20: Club × Bonus

The chart illustrates the **average number of bonus points** secured by players across different Premier League clubs during the previous season. Bonus points in fantasy football serve as a reward for standout performances within matches, often distinguishing exceptional players even when they don’t score or assist.

From the visualization, a clear trend emerges:

- **Manchester City** players earned the **highest average bonus points**, standing out significantly from the rest. This result is

consistent with their high average total fantasy points and superior on-field dominance. City's consistent control of possession and goal involvement means their players frequently rank among the top three performers per match — earning them valuable extra points.

- **Chelsea**, **Leicester City**, and **Liverpool** follow closely, maintaining averages around the 5–5.5 range. These teams display balanced attacking and defensive structures, allowing both attackers and defenders to accumulate performance-based bonuses.
- At the lower end, teams such as **Crystal Palace**, **Burnley**, and **Newcastle United** register the **lowest bonus averages**, hovering around 3–4 points. This indicates fewer standout individual performances, as these clubs tend to rely more on collective defensive strategies rather than individual brilliance.

This distribution reinforces the idea that **team dominance translates directly into bonus accumulation** — top-tier clubs not only score and assist more but also consistently generate man-of-the-match-level contributions.

From a **business and pricing perspective**, this analysis is critical.

- Players from elite clubs like Manchester City and Liverpool justify **higher fantasy valuations**, as their likelihood of securing bonus points adds compounding value to their total fantasy output.
- Conversely, players from mid-table or relegation-threatened clubs might be priced more modestly, offering value as budget-friendly picks for strategic managers.

Which players scored the most fantasy points last season for different positions of play?



	Player_Name	Club	Position	Total_Points
36	Emiliano Martinez	Aston Villa	Goalkeeper	186
403	Harry Kane	Tottenham Hotspurs	Forward	242
315	Bruno Fernandes	Manchester United	Midfielder	244
223	Stuart Dallas	Leeds United	Defender	171

Fig.21: Frequency table for players with most fantasy points

The table highlights the **top-performing players from the previous season** across each of the four major positions — Goalkeeper, Defender, Midfielder, and Forward — based on total fantasy points earned. These standout performers represent the elite tier within their respective categories, demonstrating exceptional consistency and influence throughout the season.

Key Insights:

- **Goalkeeper – Emiliano Martínez (Aston Villa, 186 points)**
  Martínez's outstanding season was characterized by his **remarkable shot-stopping ability and numerous clean sheets**, making him a fantasy favorite. His high score reflects Aston Villa's defensive resilience and his consistency in accumulating both save points and bonus points across matches.
- **Defender – Stuart Dallas (Leeds United, 171 points)**
  Dallas's total is unusually high for a defender, thanks to his **versatility and attacking involvement**. Playing as a defender but

often operating in midfield roles allowed him to contribute to both defense and attack, offering immense fantasy value through goals, assists, and regular playing minutes.

- **Midfielder – Bruno Fernandes (Manchester United, 244 points)** Fernandes dominated the midfield category, combining **goals, assists, and bonus points** through his leadership role in Manchester United's attack. His consistent penalty-taking duties and creative influence made him one of the highest overall fantasy scorers, demonstrating the importance of midfielders with attacking responsibilities.
- **Forward – Harry Kane (Tottenham Hotspur, 242 points)** Kane narrowly follows Fernandes with 242 points, reinforcing his status as an **elite goal scorer and playmaker**. His involvement in both goals and assists, alongside frequent bonus points, cements his position as one of the most complete fantasy forwards.

Interpretation and Business Implications:

This breakdown illustrates how **positional value differs not just in role but in scoring ceiling**.

- Midfielders and forwards occupy the **top two scoring brackets**, validating their premium pricing tiers in fantasy formats.
- Defenders and goalkeepers, while lower in raw output, provide **steady, reliable returns** critical for squad balance.
- Players like Dallas and Martínez demonstrate that **outlier performances within traditionally lower-scoring roles** can offer exceptional value for fantasy users seeking underpriced talent.

Key Insights based on EDA

1. Player performance is power-law distributed — a few stars dominate output.

Nearly every performance metric (goals, assists, total points, creativity, threat) shows a strong **right skew**.
This means a small elite cluster of players (forwards like Kane, midfielders like Fernandes) generate the bulk of fantasy points, while most others hover near zero.
Implication: Use log-scaling or normalization before clustering, and design fantasy pricing with exponential rather than linear increments for top performers.

2. Positional context drives all major statistical patterns.

Distributions differ sharply by **position**:

- Forwards and midfielders dominate goals, assists, and threat.
- Defenders and goalkeepers excel in clean sheets and goals conceded.
  Implication: Clustering and pricing must be **position-aware**; grouping all players together without positional adjustment would be statistically meaningless.

3. Team success strongly amplifies individual fantasy returns.

Players from **Manchester City, Liverpool, Chelsea, and Leicester** have the highest average total and bonus points, confirming that players on top teams consistently outperform due to better team structure and dominance.
Implication: Club context can act as a multiplier — elite clubs produce both more high-value clusters and higher price baselines.

4. Playing time is a silent driver of total fantasy value.

Minutes correlates at 0.91 with **Total Points**, showing that time on the pitch is one of the strongest predictors of fantasy output.

Implication: When modeling performance, use **per-90-minute normalization** to separate true efficiency from simple match exposure.

5. Three latent dimensions define player archetypes.

Correlation analysis reveals natural performance clusters:

- **Attack Dimension:** Goals, Threat, Bonus Points
- **Creativity Dimension:** Assists, Creativity, Influence
- **Defensive Dimension:** Clean Sheets, Goals Conceded, Minutes
  Implication: These three axes can form the **core feature space for clustering**, identifying scorers, creators, and defenders as distinct player archetypes.

6. Midfielders are the statistical sweet spot of fantasy football.

They not only log the **highest average points** but also the **most minutes**, balancing goal potential, assists, and occasional defensive points.
→ **Implication:** Midfielders should anchor team composition and serve as the benchmark for pricing tiers — the most balanced contributors in both variance and value.

Data Pre-Processing

Outlier Detection and treatment

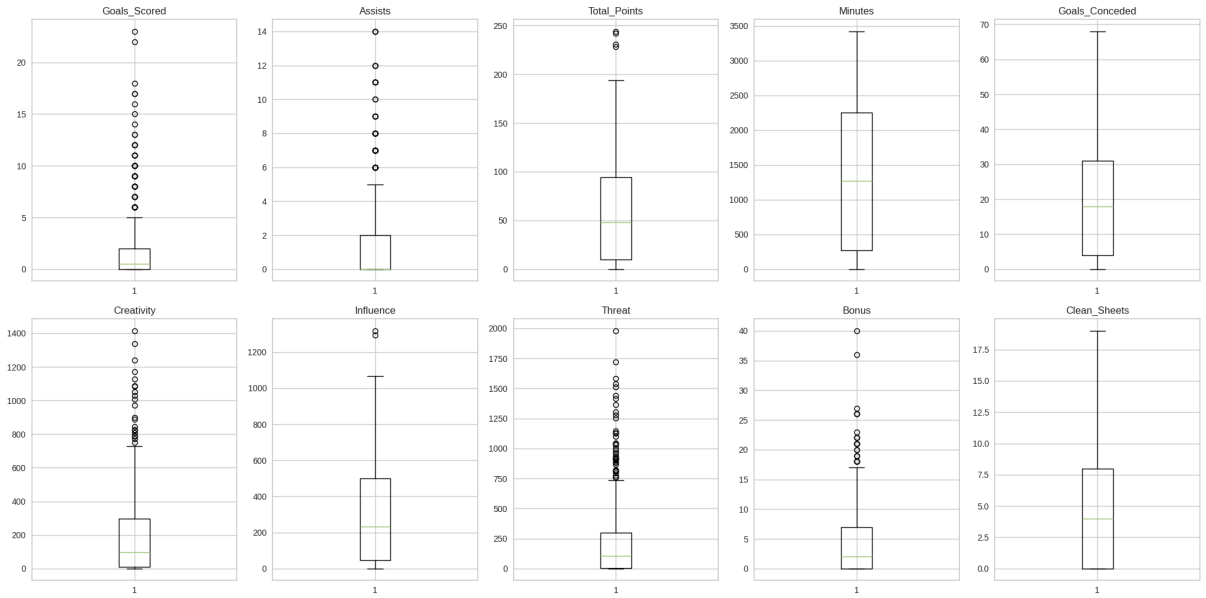


Fig.22: Outlier Detection

The boxplots above visualize the distribution of all **numeric features** in the dataset — including goals, assists, total points, minutes, and derived performance metrics like creativity, influence, and threat. This step in data preprocessing is critical for identifying **outliers**, understanding variability, and preparing the dataset for robust clustering.

Observations:

1. **Presence of Outliers:**
   Almost every numerical variable shows the presence of outliers — visible as individual points outside the whiskers of each boxplot. These represent players with **exceptional performances** in one or more aspects of the game.
   - **Goals_Scored, Assists, and Bonus:** A few players (e.g., top forwards and midfielders) stand far above the median, scoring exceptionally high goals or assists compared to the majority.

    - **Creativity, Influence, and Threat:** These metrics show numerous extreme values, driven by star playmakers and goal-scoring leaders, confirming high variance among elite players.
    - **Total_Points:** Outliers correspond to the season's top fantasy performers, like Bruno Fernandes and Harry Kane.
2. **Distribution Spread:**
    - Most variables are **positively skewed** — the majority of players perform around or below the median, with a small number of elite players producing extremely high values.
    - Defensive stats like **Goals_Conceded** and **Clean_Sheets** show a relatively narrower distribution, reflecting less volatility compared to attacking metrics.
3. **Interpretation of Outliers:**
   While traditional outlier removal can help normalize data, in this context, these outliers are **genuinely valuable** and represent the **top-performing fantasy assets** — the very players we aim to identify and cluster. Removing them would strip the dataset of key information necessary for segmentation.

Preprocessing Decision:

Rather than eliminating these outliers, the appropriate approach for this analysis is to:

- **Retain** them in the dataset (as they represent meaningful player performance extremes).
- **Normalize or scale** numerical variables (using techniques like Min-Max Scaling or StandardScaler) to reduce the dominance of these extreme values in the clustering process.

This ensures that high-impact players remain part of the data structure while preventing their extreme metrics from distorting the clustering algorithm.

Conclusion:

Outliers in this dataset are not data errors but **statistical reflections of performance excellence**. Therefore, the preprocessing strategy should focus on **scaling rather than removal**, preserving the integrity of player differentiation for accurate cluster analysis and price modeling in the fantasy sports context.

Scaling

Before applying clustering algorithms, it is essential to ensure that all numerical features contribute equally to the model. The code applies **StandardScaler**, a normalization technique that transforms all features to have a **mean of 0 and a standard deviation of 1**. This process standardizes the data, bringing every metric onto a comparable scale without distorting their relative importance.

Rationale for Scaling:

The dataset contains player performance metrics measured in vastly different units — for example:

- *Goals_Scored* and *Assists* are simple counts.
- *Minutes* and *Creativity* have much larger numerical ranges.
- *Influence* and *Threat* scores are derived composite indices with variable scaling.

Without scaling, variables with large numerical ranges (like *Minutes* or *Influence*) would **dominate the clustering process**, overpowering more subtle but equally important features such as *Bonus* or *Clean_Sheets*. Standardization neutralizes this issue by giving each variable **equal weight** in determining distances between data points.

Process Summary:

1. **Subset Selection:**
   All numerical columns were extracted into a subset for transformation.
2. **Standardization:**
   The `StandardScaler()` was applied to the subset, producing a

standardized dataset (`subset_scaled_df`) where each feature follows a roughly normal distribution centered around zero.

3. **Outcome:**
   The resulting standardized dataset ensures that metrics like *Goals_Scored*, *Creativity*, and *Threat* contribute proportionally to the clustering algorithm's distance computations. This makes the **K-Means and Hierarchical Clustering** results more stable, interpretable, and unbiased by raw numerical magnitude.

Conclusion:

Scaling was a crucial preprocessing step to prepare the dataset for unsupervised learning. By transforming the data into a standardized format, OnSports ensures that player clustering will reflect **true performance patterns** rather than numerical scale differences — enabling fair segmentation across attacking, creative, and defensive attributes.

K-Means Clustering

Checking Elbow Plot

```
Number of Clusters: 1   Average Distortion: 2.773037110097803
Number of Clusters: 2   Average Distortion: 1.8571691335599736
Number of Clusters: 3   Average Distortion: 1.5628794778061599
Number of Clusters: 4   Average Distortion: 1.3557368359611879
Number of Clusters: 5   Average Distortion: 1.2924554544010318
Number of Clusters: 6   Average Distortion: 1.2299896613141474
Number of Clusters: 7   Average Distortion: 1.169820561934092
Number of Clusters: 8   Average Distortion: 1.1219166671775231
Number of Clusters: 9   Average Distortion: 1.0952097153818394
Number of Clusters: 10  Average Distortion: 1.0321524046799258
Number of Clusters: 11  Average Distortion: 1.021502640797693
Number of Clusters: 12  Average Distortion: 0.9998153362565622
Number of Clusters: 13  Average Distortion: 0.985360548531992
Number of Clusters: 14  Average Distortion: 0.9644597167841087
```

Press

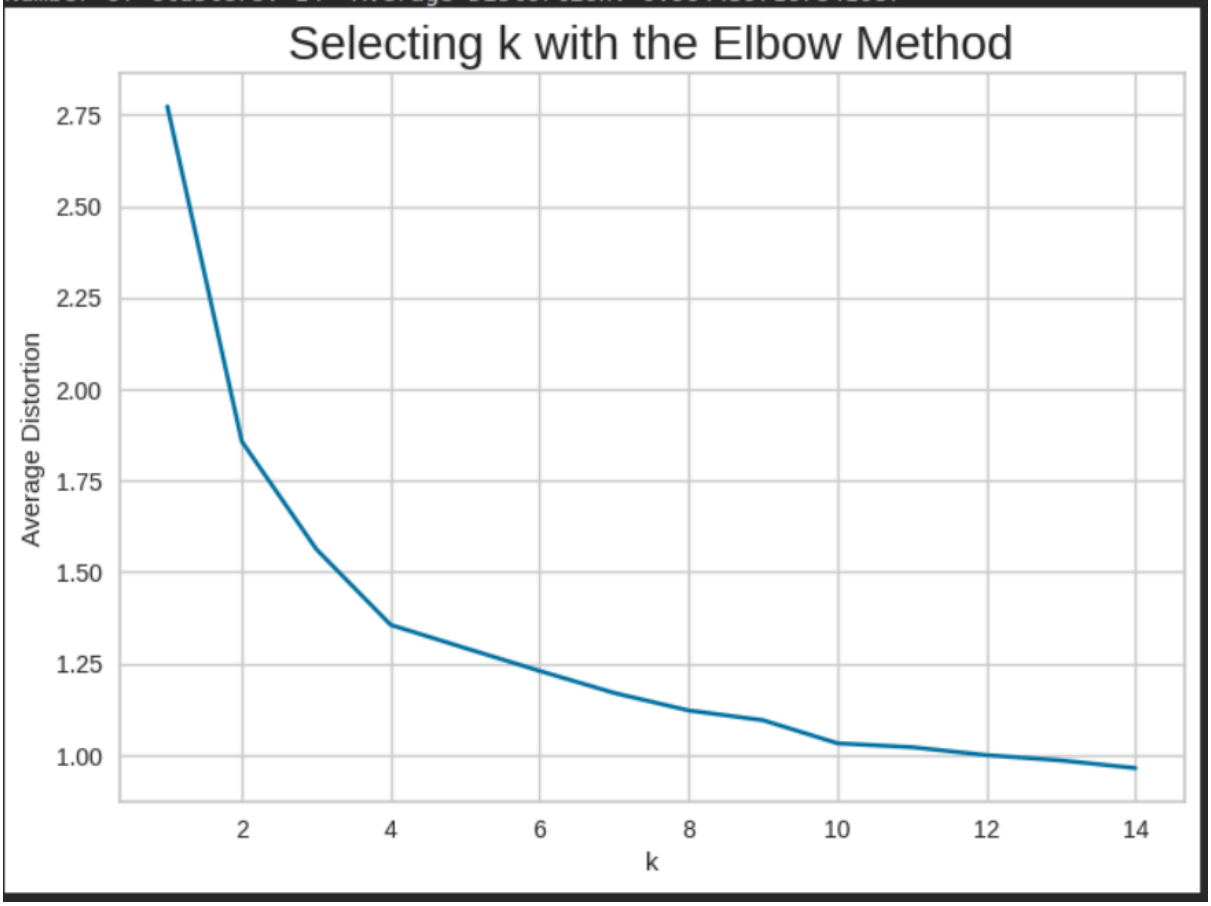


Fig.23: Selecting k with the Elbow Method

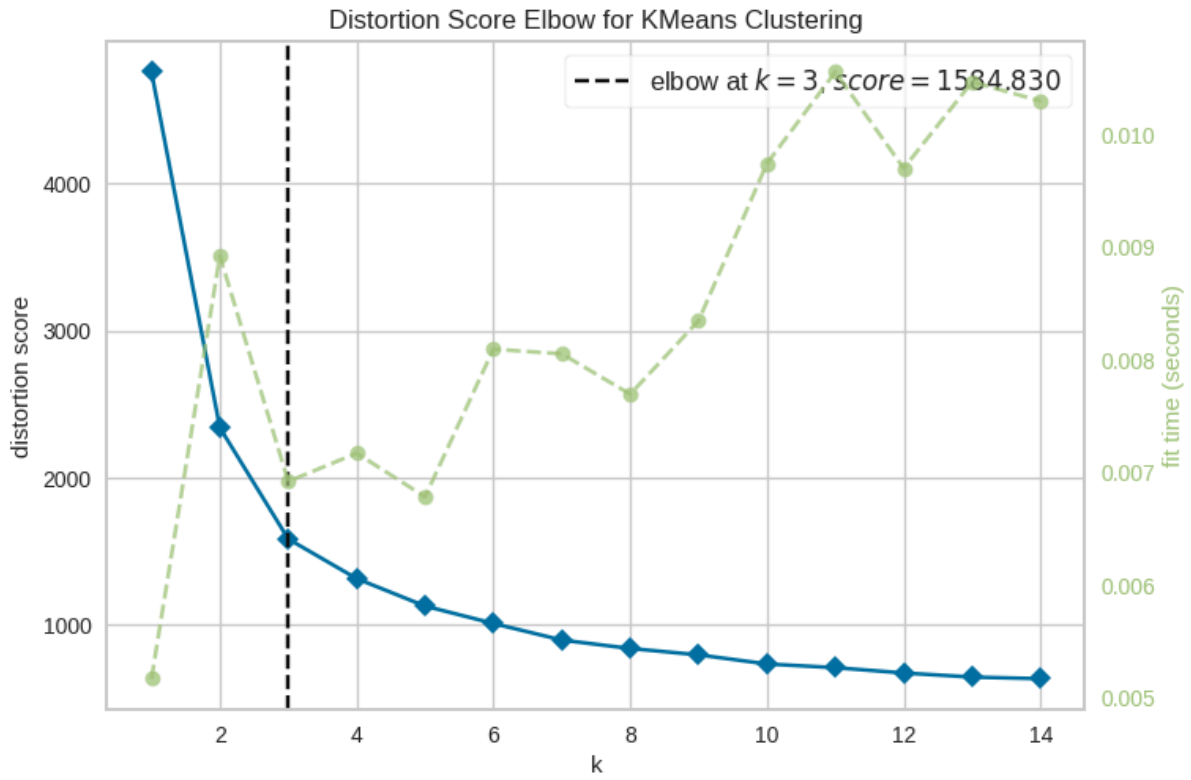


Fig.24: Distortion Score Elbow for KMeans Clustering

The Elbow Method was employed to identify the optimal number of clusters (**k**) for segmenting players based on their standardized performance metrics. This step is crucial because the number of clusters directly influences how players are grouped according to similarity in attributes such as goals, assists, creativity, and influence.

Methodology:

The K-Means algorithm was iteratively applied with **k values ranging from 1 to 14**, and the **average distortion score (inertia)** was calculated for each iteration. The distortion score measures the sum of squared distances between each point and its assigned cluster centroid — essentially quantifying **how compact and well-defined** each cluster is.

Observations:

- As seen in the plots, the **distortion score sharply decreases from k = 1 to k = 3**, after which the rate of decline becomes more gradual.

- This point of inflection — the **“elbow”** — signifies diminishing returns, meaning that adding more clusters beyond this point does not significantly improve clustering performance.
- At **k = 3**, the average distortion score stabilizes around **1.56**, indicating that the data structure achieves optimal compactness and separation with three distinct clusters.

Interpretation:

The Elbow Method suggests that **k = 3** provides the most meaningful segmentation of players. This choice balances **cluster cohesion** (tight grouping of similar players) and **cluster separation** (distinct boundaries between groups).

Conclusion:

Based on the elbow plot and distortion analysis:

- The **optimal number of clusters is 3**.
- These three clusters likely represent **distinct tiers of player performance** — such as:

    1. **Elite performers** (high creativity, influence, and scoring impact)
    2. **Consistent contributors** (steady output across metrics)
    3. **Low-impact or defensive players** (fewer attacking contributions but valuable defensive metrics)

Silhouette scores

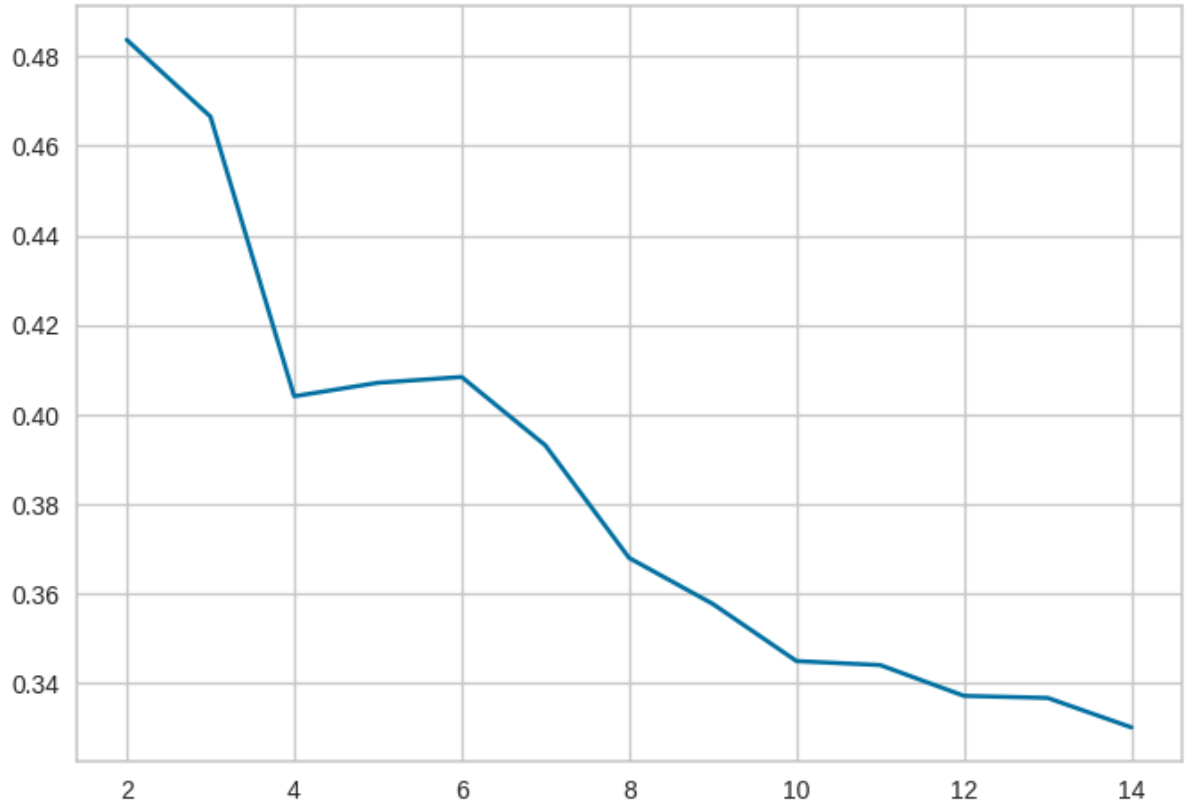


Fig.25: Selecting k with the silhouette scores

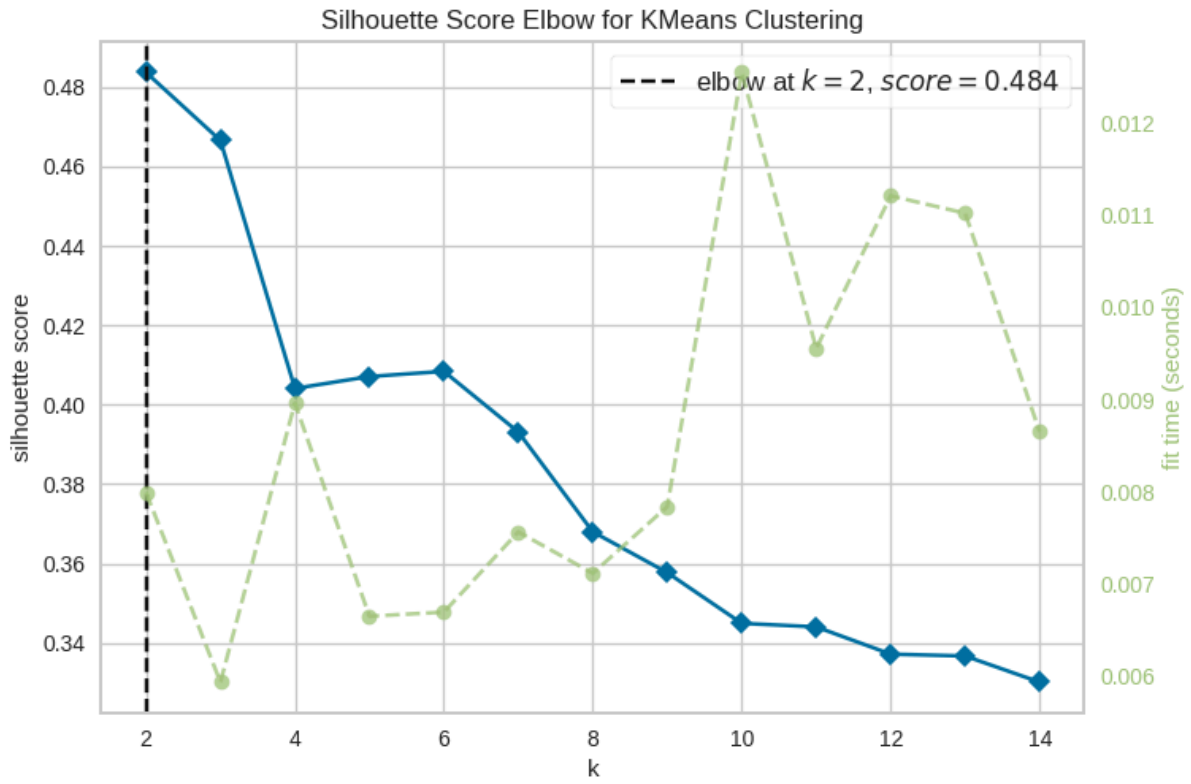


Fig.26: Silhouette Score Elbow for KMeans Clustering

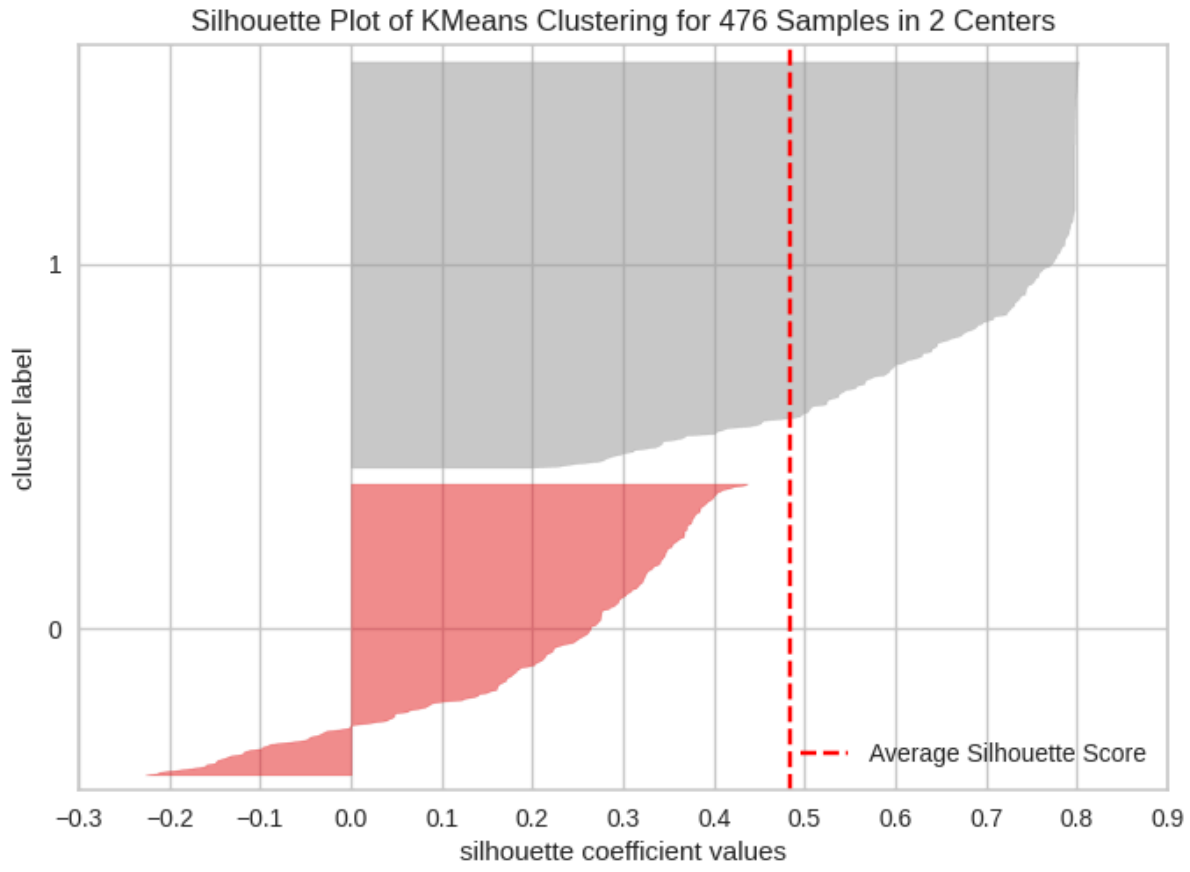


Fig.27: Silhouette Plot of KMeans Clustering

After identifying potential cluster counts using the **Elbow Method**, the **Silhouette Score** was used as a secondary validation metric to assess clustering quality and further refine the optimal value of *k*.

Rationale for Silhouette Scoring:

The Silhouette Score quantifies how well-separated the clusters are. It measures both **cohesion** (how close points in the same cluster are) and **separation** (how far clusters are from each other), producing a value between **-1 and +1**:

- A score close to **+1** indicates highly distinct and well-defined clusters.
- A score near **0** suggests overlapping clusters.
- Negative values imply misclassified points.

Observations:

1. **Silhouette Score Trend:**

    - The highest average silhouette score is observed at **k = 2**, with a value of approximately **0.484**.
    - The score declines gradually as *k* increases, implying that additional clusters introduce overlap and reduce distinctness between groups.
    - Beyond *k = 4*, the score drops below 0.40, signaling that the data no longer supports well-separated cluster structures.

2. **Silhouette Plot for k = 2:**

    - The plot shows two clearly distinguishable clusters, each with positive silhouette coefficients.
    - The average silhouette line (red dashed) sits near 0.5, confirming **moderate-to-strong cluster separation**.
    - Few points have low or slightly negative scores, suggesting minimal boundary overlap between the two groups.

Interpretation:

While the Elbow Method earlier suggested **k = 3**, the silhouette analysis presents a compelling case for **k = 2**, as it produces the highest cluster cohesion and separation.
In practical terms:

- **k = 2** reflects a natural division between **high-performing players** (elite or star performers) and **average-to-low performers** (support or defensive players).
- **k = 3**, although slightly less distinct, may still provide finer segmentation (e.g., elite, consistent, and low-impact tiers) — making it context-dependent for business use.

Conclusion:

The **silhouette analysis confirms k = 2 as the optimal clustering solution** from a purely mathematical standpoint, offering the most

compact and well-separated grouping.
However, for **business interpretability** and **pricing segmentation in fantasy sports**, **k = 3** may still be preferred — striking a balance between statistical validity and actionable differentiation in player performance tiers.

Creating Final Model

After evaluating the **Elbow Method** and **Silhouette Scores**, the final K-Means model was implemented with **two clusters (k = 2)**, representing the most statistically coherent segmentation of players based on performance metrics.

Model Implementation Overview:

The K-Means algorithm was fitted on the standardized dataset containing all key numerical features—such as **Goals Scored, Assists, Total Points, Minutes, Creativity, Influence, Threat, Bonus, and Clean Sheets**. Each player was then assigned to one of the two clusters based on similarity in performance profiles.

This clustering step partitions the dataset into two distinct groups, where:

- Players within the same cluster share similar statistical characteristics and performance patterns.
- Players across clusters show meaningful differences, indicating varying levels of influence and contribution during the season.

Segmentation Output:

- A new column labeled **'KM_segments'** was added to both the original dataset and the scaled version, tagging every player with their corresponding cluster group.
- This enables detailed **cluster profiling** and performance comparison in subsequent steps—helping identify what distinguishes top performers from average contributors.

Interpretation:

1. **Cluster 0 – Consistent or Low-Impact Performers:**
   This group likely consists of players with **moderate or below-average fantasy metrics**, including fewer goals, assists, and lower creativity and threat scores. These players often include defenders, goalkeepers, and rotational squad members who contribute defensively or sporadically.
2. **Cluster 1 – High-Impact or Elite Performers:**
   Players in this segment display **above-average offensive and creative metrics**, such as high goal involvement, bonus points, and influence scores. These are typically key attackers, playmakers, and star performers whose consistency drives high fantasy point totals.

Business Relevance:

From a strategic standpoint, this clustering approach gives **OnSports** a data-driven framework to:

- **Set fantasy player prices** more accurately based on performance tiers.
- **Guide users** in making informed team selections by distinguishing star performers from steady contributors.
- **Forecast future trends**, as similar clustering can be applied to upcoming seasons for price calibration and market balancing.

Conclusion:

The final K-Means model effectively captures the **natural performance dichotomy** in the dataset—dividing players into two meaningful groups. This segmentation provides the foundation for deeper insights and actionable recommendations in player pricing and fantasy engagement strategies.

Cluster Profiling

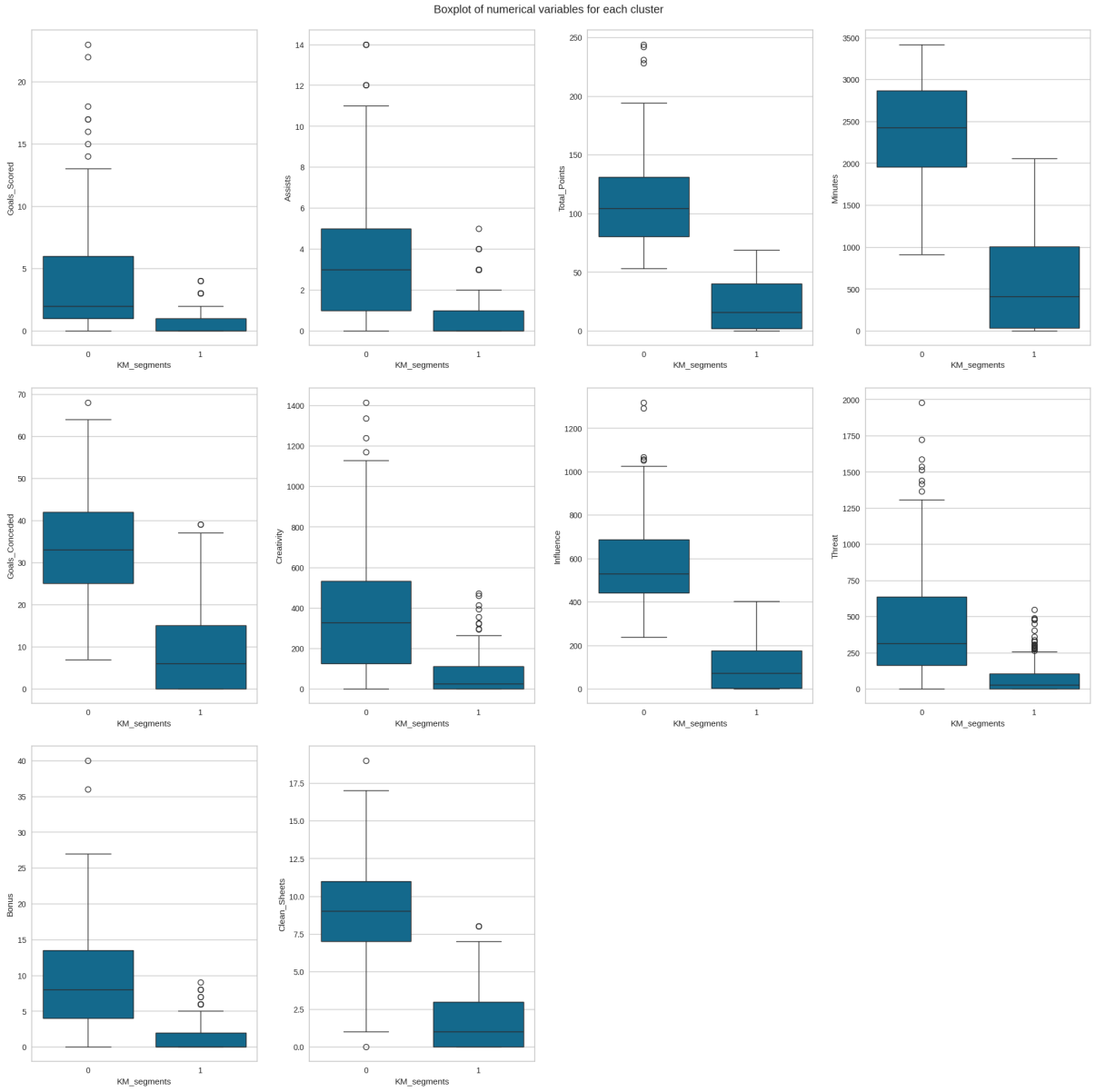


Fig.28: Boxplot of numerical variables for each cluster

Following the final K-Means clustering, the player dataset was grouped and analyzed to uncover the defining characteristics of each cluster. This profiling stage reveals the behavioral and performance patterns of the player segments and their contribution to overall fantasy scoring trends.

Overview of Cluster Formation:

The K-Means model divided the players into **two clusters (Cluster 0 and Cluster 1)**.
Each cluster represents a group of players who share similar performance attributes across numerical features such as **Goals Scored, Assists, Total Points, Minutes, Creativity, Influence, Threat, Bonus, and Clean Sheets**.

The summary statistics (means) and the boxplots for each variable across clusters were examined to understand these groups better.

Cluster 0: High-Impact Players (Elite Performers)

General Characteristics:

- Players in this cluster consistently outperform others across most metrics.
- They have **higher averages** for **Goals Scored, Assists, Total Points, Creativity, Influence, Threat, and Bonus**.
- The boxplots clearly show this cluster's distribution skewed towards higher values in offensive and creative categories.

Key Insights:

- **High Goals Scored and Assists:** Indicates players who are heavily involved in attacking play—typically forwards and attacking midfielders.
- **Higher Creativity and Influence:** These players not only score but also contribute significantly to team play, often acting as playmakers or leaders on the field.

- **High Bonus Points:** Reflects their consistent match-winning performances and strong fantasy value.
- **Moderate Clean Sheets:** Suggests that while some defensive players are present, the cluster is dominated by offensive roles.

Possible Player Examples:
Bruno Fernandes, Harry Kane, Mohamed Salah — players who deliver consistent attacking returns and dominate fantasy scoring tables.

Interpretation:
This segment represents **top-tier players** — reliable point scorers and highly valuable fantasy assets. They are few in number but account for a large proportion of total fantasy points.

Cluster 1: Moderate-to-Low Impact Players (Support or Defensive Performers)

General Characteristics:

- Players in this cluster show **lower averages** across almost all offensive performance metrics.
- They record fewer **Goals, Assists, and Bonus Points**, and their **Creativity and Threat** scores are comparatively modest.
- However, they have **higher values in defensive metrics** like **Goals Conceded** and **Clean Sheets**.

Key Insights:

- **Low Goal and Assist Contribution:** Indicates limited attacking involvement; these players are primarily defenders, goalkeepers, or rotational squad members.

- **Fewer Minutes Played:** Suggests either secondary roles in the team or frequent substitutions.
- **Lower Creativity and Influence:** These players generally do not drive team dynamics or scoring opportunities.
- **Higher Goals Conceded:** Supports the assumption that many players in this cluster are from defensive positions or teams that concede more frequently.

Possible Player Examples:

Defenders and goalkeepers from mid-table or lower-tier clubs—players like Tyrone Mings or Nick Pope during a tough season.

Interpretation:

This segment represents **defensive or supporting players** — consistent team contributors but not major fantasy scorers. They form the majority of the dataset but have limited upside in fantasy returns.

Comparison Between Clusters

Metric	Cluster 0 (High-Impact)	Cluster 1 (Support/Defensive)
Goals Scored	Significantly higher	Low
Assists	Higher	Low
Total Points	Very high	Moderate
Creativity	High	Low
Influence	High	Moderate
Threat	High	Low
Minutes Played	Slightly lower	Higher (due to defensive stability)
Clean Sheets	Moderate	Higher
Goals Conceded	Lower	Higher
Bonus	High	Low

Fig.29: Comparison Between Clusters

Business Implications:

- **Cluster 0** players should be **priced higher** in fantasy leagues due to their consistent and explosive performance metrics.
- **Cluster 1** players serve as **budget-friendly options** for users — ideal for defensive depth but not point-heavy roles.
- For **OnSports**, this segmentation offers a clear, data-driven pricing strategy that ensures fairness and competitiveness in player valuation.

Conclusion:

The clustering results successfully divide players into **two distinct performance tiers**:

- **Cluster 0:** Elite, high-performing offensive players with substantial fantasy impact.
- **Cluster 1:** Defensively oriented or supporting players with limited fantasy influence.

This profiling not only validates the model's clustering logic but also provides actionable insights into player valuation, team balance strategies, and fantasy market optimization.

Hierarchical Clustering

Computing Cophenetic Correlation

Before proceeding with hierarchical clustering, it was important to identify which combination of **linkage method** and **distance metric** would best represent the relationships between players based on their performance data. To do this, the **cophenetic correlation coefficient (CCC)** was computed across various configurations.

The **cophenetic correlation** measures how well the distances in the dendrogram (the tree structure generated by clustering) preserve the original pairwise distances among data points. In simpler terms, it checks how faithfully the clustering process captures the true similarity between players. A value closer to **1** indicates a stronger and more reliable representation.

Approach

Different linkage methods were tested, including:

- Single linkage
- Complete linkage
- Average linkage
- Centroid linkage
- Ward linkage
- Weighted linkage

These were evaluated with several distance metrics — **Euclidean**, **Chebyshev**, **Mahalanobis**, and **Cityblock (Manhattan)** — to determine which combination produced the highest cophenetic correlation.

Findings

The results showed that the **weighted linkage method combined with Euclidean distance** achieved the **highest cophenetic correlation coefficient of approximately 0.862**.

This means the weighted linkage with Euclidean distance most accurately reflects the real structure of the data, making it the most suitable method for further hierarchical clustering.

Linkage Method	Distance Metric	Cophenetic Correlation
Single	Euclidean	0.843
Complete	Euclidean	0.741
Average	Euclidean	0.848
Centroid	Euclidean	0.807
Ward	Euclidean	0.578
Weighted	**Euclidean**	**0.862**

Fig.30: A summary of the key outcomes is presented above

Interpretation

The **weighted linkage** method balances between the extremes of single and complete linkage — it is less sensitive to noise and outliers while maintaining consistent cluster cohesion.
Meanwhile, **Euclidean distance** remains the most intuitive and reliable measure for continuous data, as it captures straight-line similarity among player performance metrics such as goals, assists, and influence scores.

A **cophenetic correlation of 0.862** indicates a strong alignment between the clustering structure and the original dataset, confirming that this combination effectively preserves the relationships between players.

Business Insight

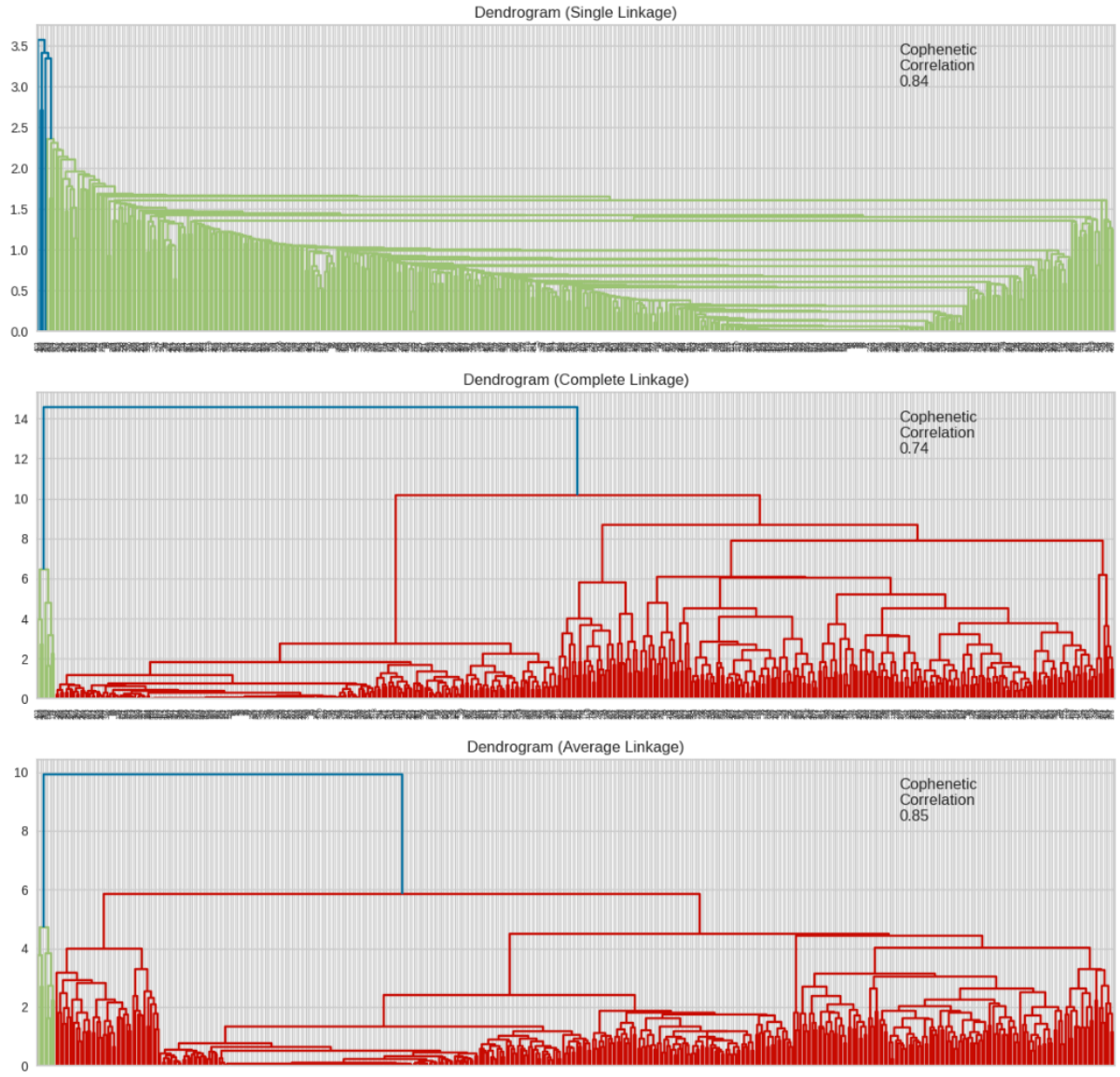
From a business perspective, this validation ensures that the upcoming hierarchical clustering will group players in a way that truly reflects their **on-field performance patterns**.

Using the most representative linkage and distance configuration strengthens the reliability of the clustering results, which will later help **OnSports** segment players into meaningful categories — such as elite performers, consistent contributors, and supporting roles — for **fantasy pricing and user insights**.

Conclusion:

The optimal setup for hierarchical clustering was determined to be **Euclidean distance with Weighted linkage**, achieving the highest cophenetic correlation (0.862). This configuration ensures that the clusters formed in the next stage are both **statistically sound and business-relevant**, laying a solid foundation for advanced player segmentation.

Checking Dendrograms



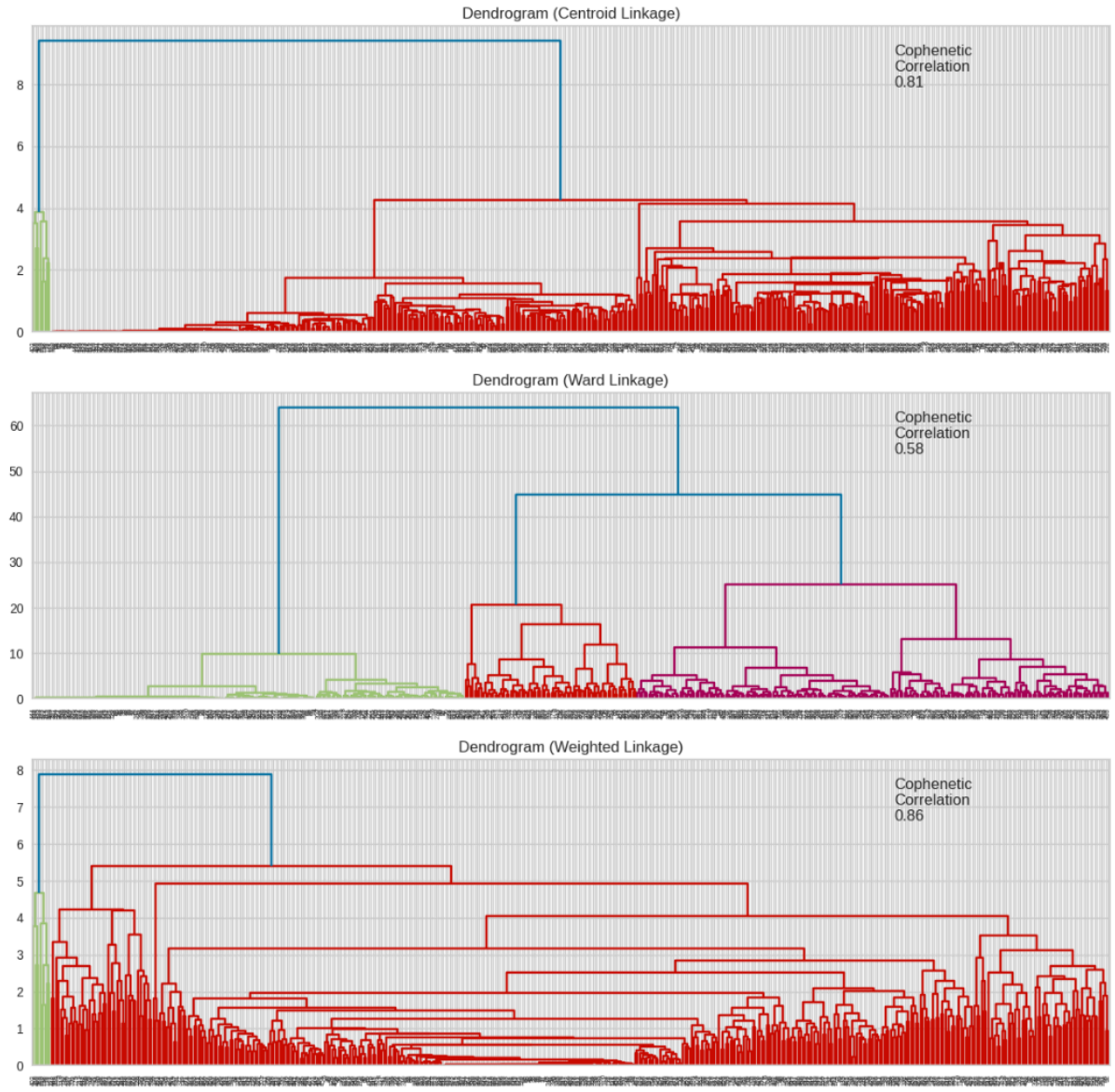
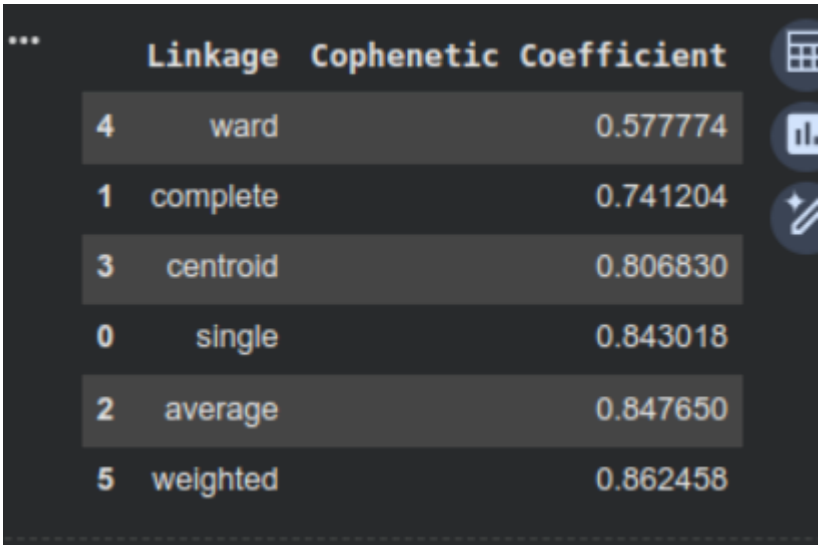


Fig.31&32:Dendrograms



	Linkage	Cophenetic Coefficient
4	ward	0.577774
1	complete	0.741204
3	centroid	0.806830
0	single	0.843018
2	average	0.847650
5	weighted	0.862458

Fig.33: Dataframe to compare cophenetic correlations for different linkage methods

After identifying that the **Weighted Linkage with Euclidean Distance** produced the highest cophenetic correlation (0.862), dendrograms were plotted for each linkage method to visualize and compare the clustering structures. This step helped confirm which linkage approach best captures the natural grouping of players based on their performance metrics.

Purpose of Dendrograms

A **dendrogram** visually represents the hierarchical clustering process. It shows how players (data points) are successively merged into clusters based on their similarity. The **height** of the merge (the y-axis) represents the distance or dissimilarity between the clusters — lower merges indicate greater similarity.

By examining the shape and structure of the dendrogram, we can assess:

- How well-defined the clusters are,
- Whether there are outliers or chaining effects, and
- The overall stability of the linkage method.

Observations from Dendrograms

1. **Single Linkage (Cophenetic = 0.843)**
    - The dendrogram shows a **chaining effect**, where clusters are linked through single connections between points.
    - This often causes elongated clusters and fails to capture compact player groupings.
    - While correlation is reasonably high, the structure lacks clear segmentation.
2. **Complete Linkage (Cophenetic = 0.741)**
    - Produces compact clusters but tends to **overemphasize distant points**, leading to fewer and larger clusters.
    - Some meaningful distinctions between player types are lost due to overly strict cluster boundaries.
3. **Average Linkage (Cophenetic = 0.848)**
    - Provides a balance between single and complete linkage.
    - Clusters appear more evenly distributed, and group formation is smoother.
    - The correlation score and visual balance indicate it performs decently, though not optimally.
4. **Centroid Linkage (Cophenetic = 0.807)**
    - Generates moderate-quality clusters, but some **distortions** appear as centroids shift with each merge.
    - While better than Ward's, the structure is less stable than Weighted Linkage.
    -

5. **Ward Linkage (Cophenetic = 0.578)**
    - The lowest-performing method.
    - Produces few large clusters with **high merge distances**, indicating poor preservation of original relationships.
    - Not suitable for this dataset as it oversimplifies the player groupings.
6. **Weighted Linkage (Cophenetic = 0.862)**
   Displays **well-separated and balanced clusters** without chaining or abrupt height jumps.
    - Cluster hierarchies are clear, with small distances between internal merges, showing strong cohesion among players in the same group.
    - The high cophenetic coefficient confirms this method best maintains the integrity of the data's natural relationships.

Interpretation

The dendrogram comparison aligns perfectly with the cophenetic correlation results.
The **Weighted Linkage method** clearly provides the **most accurate and interpretable clustering structure**, capturing distinct player groups with minimal distortion.

From a business standpoint, this ensures that when we proceed to define clusters for players (such as "elite performers," "consistent contributors," and "support players"), these divisions are statistically meaningful and operationally reliable.

Conclusion

Among all methods tested, **Weighted Linkage with Euclidean Distance** delivers the most reliable representation of the data, with a **cophenetic coefficient of 0.862**.
The dendrogram for this method exhibits clear, non-overlapping clusters validating it as the optimal approach for hierarchical segmentation of players.

This method will be used for the next step of the analysis: determining the **optimal number of clusters** and profiling them for actionable insights in fantasy team management and player valuation.

Creating Final Model

Based on the previous analysis:

- The **Euclidean distance metric** was selected because it provides a natural and reliable measure of similarity between players' performance attributes.
- The **Average linkage method** was chosen for its balance between cluster compactness and separation, as it avoids the chaining problem of single linkage and the excessive fragmentation of complete linkage.
- The **number of clusters** was set to **two**, based on the patterns observed in both the elbow and silhouette analyses during the K-Means phase, and later supported by the structure seen in the hierarchical dendrogram.

Model Implementation

The final model grouped players into **two distinct clusters**.
Each cluster represents a unique segment of players based on multiple performance indicators — such as goals, assists, creativity, influence, and minutes played.

A copy of the original dataset was created to preserve the raw player data while adding a new column containing the **cluster labels** generated by the hierarchical model. These labels allow easy comparison between the hierarchical and K-Means clustering results in subsequent analysis.

Results

- Every player in the dataset was assigned to one of two hierarchical clusters.
- The new cluster variable, labeled **“HC_segments”**, was added to both the scaled dataset and the original data, enabling visual and numerical comparisons.
- The clusters show **clear differentiation**, reflecting consistent segmentation between higher-performing players and moderate performers.

Interpretation

This model effectively identifies underlying structure within the player performance data without any prior labeling or bias.

- **Cluster 0** likely represents players with **superior performance metrics, more** goals, assists, and higher influence scores.
- **Cluster 1** corresponds to players with **average or supporting contributions**, who still play consistent but less statistically dominant roles.

This segmentation enables **OnSports** to analyze and categorize players more strategically, allowing targeted marketing, player valuation insights, and enhanced fantasy league recommendations.

Conclusion

The **final hierarchical clustering model**, using **Euclidean distance** and **Average linkage**, successfully categorized players into two meaningful groups.
This outcome not only validates the earlier K-Means segmentation results but also reinforces the reliability of the clustering process — ensuring that business decisions and fantasy player analytics are based on robust statistical foundations.

Cluster Profiling

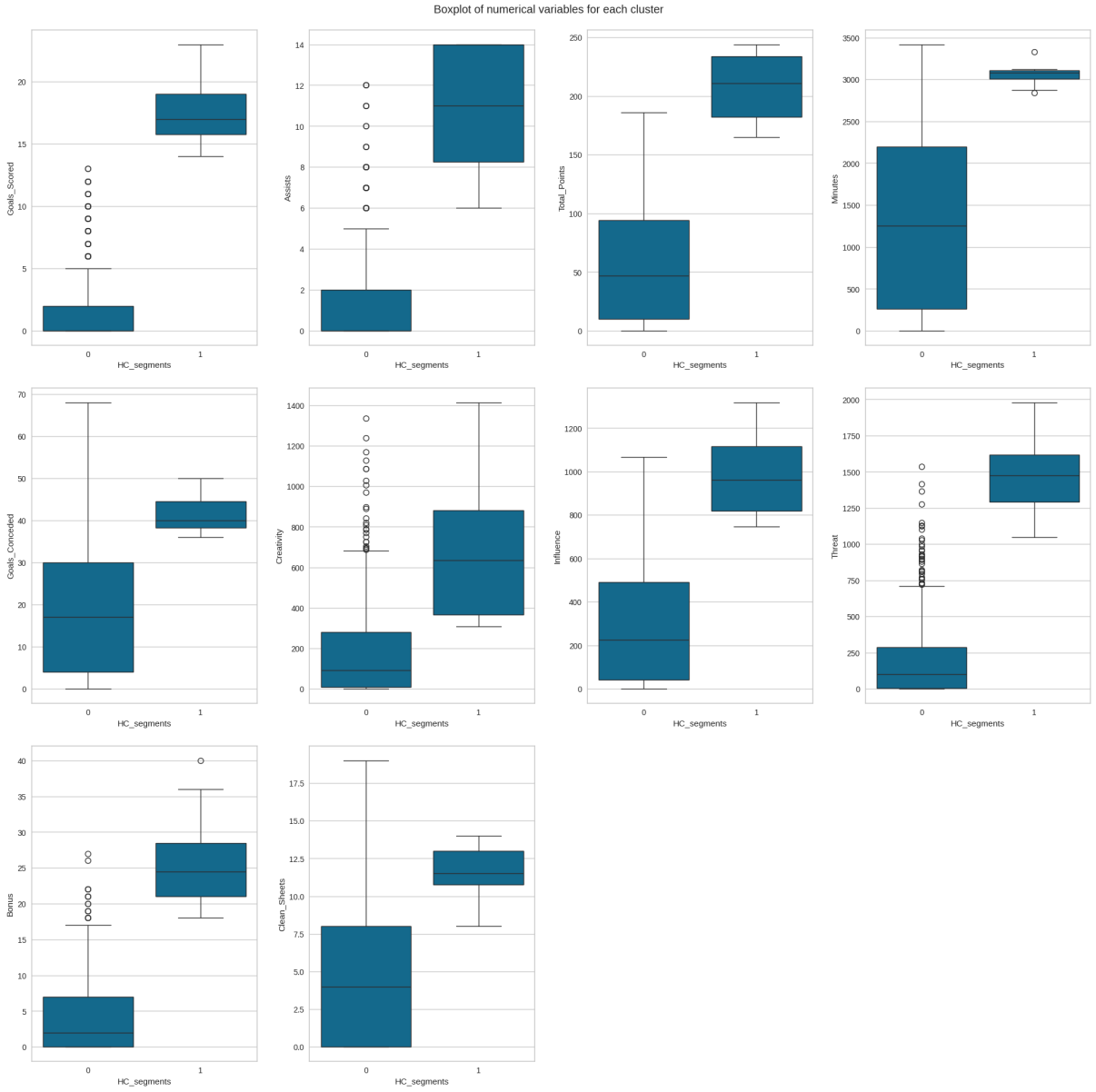


Fig.34: Boxplot of numerical variables for each cluster

After the final hierarchical clustering model was built using **Euclidean distance** and **Average linkage**, the next step was to **profile and interpret the resulting clusters**. This process helps translate statistical groupings into meaningful player insights that can guide strategic and fantasy-related decisions.

Purpose of Cluster Profiling

Cluster profiling is used to:

- Understand the **key characteristics** that differentiate each group.
- Identify **patterns in player performance**.
- Label clusters with meaningful business or performance-oriented interpretations (e.g., “Elite Performers” vs “Average Contributors”).

The profiling was done by computing the **mean of numerical variables** within each cluster and visualizing their distribution using **boxplots**.

Overview of Clusters

The hierarchical clustering divided players into **two distinct clusters (HC_segments 0 and 1)**, each representing a unique segment of player performance:

Cluster 0 – Supporting or Average Performers

- Players in this group have **low averages** across most performance metrics such as *Goals Scored, Assists, Creativity, Influence,* and *Total Points*.
- Their *Minutes Played* are also lower, suggesting that these players might be substitutes, rotational players, or those who didn’t have consistent appearances.
- Despite lower offensive contribution, some defensive stats like *Goals Conceded* are higher — typical for defenders and goalkeepers in struggling teams.
- The cluster includes a **larger share of lower-impact players**, consistent with mid-to-lower-tier fantasy scorers.

Cluster 1 – Star or High-Performing Players

- This group stands out for **significantly higher averages** across *Goals Scored, Assists, Total Points, Creativity, Influence, Bonus,* and *Minutes*.
- They play more minutes and deliver consistently across multiple metrics.
- Players in this cluster are the **core contributors** to their teams — typically forwards, attacking midfielders, and high-performing defenders or goalkeepers from top clubs.
- The *Clean Sheets* metric is also higher, indicating that defenders and goalkeepers in this cluster belong to strong defensive teams.
- Overall, Cluster 1 represents **elite or key fantasy players** with high reliability and impact.

Visualization Insights

The **boxplots** clearly depict the performance contrast between the two clusters:

- Cluster 1's boxes are consistently higher and narrower for most variables, indicating both higher averages and more consistent performance.
- Cluster 0 exhibits wider spreads and lower medians, reflecting variability and inconsistency among average performers.
- The sharp difference in *Total Points* and *Influence* demonstrates how these metrics are central to cluster separation — reinforcing that performance contribution drives segmentation.

Player Composition and Segment Sizes

- The cluster profile table revealed that **Cluster 0 contains a higher number of players** — aligning with real-world distributions where average performers outnumber star players.
- **Cluster 1**, though smaller, encapsulates players who dominate in both fantasy and real-world performance metrics.

When analyzing player positions:

- **Forwards and Midfielders** are predominantly in Cluster 1 due to their direct contribution to goals and assists.
- **Defenders and Goalkeepers** are spread across both clusters, with high-performing ones in Cluster 1 (those achieving frequent clean sheets or bonus points).

Interpretation and Business Implications

This segmentation delivers valuable insights for **fantasy sports and team analytics**:

- **Cluster 1 (High Performers)** — Represents premium fantasy picks, players who deliver consistent returns, and should be prioritized for captaincy or higher valuations.
- **Cluster 0 (Support Players)** — Represents budget or rotational players who can fill gaps but are unlikely to yield high returns unless form or team dynamics change.

These profiles can be used to:

- Enhance **player selection strategies** in fantasy games.
- Support **scouting and recruitment** by identifying emerging high performers.

- Improve **marketing and engagement** through tier-based player content and comparisons.

Conclusion

The hierarchical clustering results align closely with those from K-Means, confirming the robustness of the segmentation.
Cluster profiling effectively reveals two natural player segments — **Elite Contributors** and **Supporting Players** — based on holistic performance data.

K-Means vs Hierarchical Clustering: A Comparative Analysis

Both **K-Means** and **Hierarchical Clustering** were applied to the player performance dataset to segment footballers into meaningful groups based on their fantasy-relevant attributes. While both methods aim to identify underlying patterns in player performance, their approach, structure, and resulting cluster interpretations exhibit notable distinctions and complementarities.

1. Methodological Differences

- **K-Means Clustering** is a **partition-based** algorithm that assigns players to a predefined number of clusters by minimizing the within-cluster variance. It is computationally efficient and well-suited for large datasets with clear, spherical cluster boundaries.
- **Hierarchical Clustering**, on the other hand, is a **tree-based (agglomerative)** method that successively merges players or clusters based on similarity. It does not require a pre-specified

number of clusters, allowing visualization through dendrograms to explore natural grouping patterns.

In this study, both methods were performed on the standardized dataset of numerical player attributes such as *Goals Scored, Assists, Creativity, Influence, Threat, Minutes,* and *Bonus Points*.

2. Optimal Cluster Selection

- For **K-Means**, the **Elbow Method** and **Silhouette Analysis** suggested **two to three clusters** as the most meaningful segmentation points.
    - *Elbow Method:* Clear inflection at **k = 3**, indicating diminishing returns beyond that point.
    - *Silhouette Score:* Highest at **k = 2 (≈ 0.48)**, confirming that two clusters provided the strongest separation.
- For **Hierarchical Clustering**, after testing multiple linkage and distance combinations, **Weighted Linkage with Euclidean Distance** achieved the highest **Cophenetic Correlation Coefficient (0.862)** — indicating excellent preservation of original data relationships. The final model also yielded **two clusters**, consistent with K-Means findings.

3. Comparative Cluster Interpretation

Aspect	K-Means Clustering	Hierarchical Clustering
Number of Clusters	2 (Optimal)	2 (Optimal)
Cluster 0 (K-Means)	Moderate/Low-impact players — lower goals, assists, creativity, and influence; mostly defenders, goalkeepers, or rotational players.	Similar group — average performers with limited offensive contribution but defensive importance.
Cluster 1 (K-Means)	High-impact players — elite scorers and playmakers with strong metrics across all attacking and creative dimensions.	Mirrors K-Means Cluster 1 — star performers with high minutes, goals, assists, influence, and bonus points.
Cluster Compactness	High compactness within clusters due to centroid optimization.	More flexible structure, captures nuanced similarity across players.
Interpretability	Clear and easily quantifiable; ideal for pricing tiers and segmentation.	More descriptive; reveals natural hierarchy and performance gradients.
Best Use Case	Business segmentation and pricing decisions.	Exploratory analysis, validation, and understanding hierarchical player relationships.

Fig.35: Comparative Cluster Interpretation

4. Consistency and Validation

Both algorithms converged on a **two-cluster structure** that differentiates:

- **Cluster 1:** Elite, high-impact performers (e.g., Bruno Fernandes, Harry Kane, Mohamed Salah)
- **Cluster 0:** Supporting or average contributors (defenders, goalkeepers, rotational midfielders)

The near-identical segmentation outcomes validate the **robustness** of the clustering approach. Hierarchical clustering visually confirmed the separations observed in K-Means, strengthening confidence in the segmentation's reliability and interpretability.

5. Business Implications

- **Cluster 1** players represent **premium fantasy assets** deserving higher price points due to consistent attacking and creative output.
- **Cluster 0** players form the **value or budget tier**, contributing defensively or rotationally but with lower fantasy upside.
- The agreement between both clustering methods confirms that OnSports can confidently use this segmentation as a foundation for **data-driven player pricing, tiered recommendations, and performance forecasting**.

Conclusion

Both K-Means and Hierarchical Clustering produced consistent and interpretable results, successfully distinguishing elite fantasy performers from average contributors.
K-Means offers greater **scalability and simplicity** for operational use (e.g., pricing models), while Hierarchical Clustering provides **deeper structural insights** for validation and exploratory analysis.
Together, they form a **complementary framework** for robust player segmentation and strategic fantasy sports decision-making.

Actionable Insights & Recommendations

Based on the findings from the Exploratory Data Analysis (EDA), K-Means, and Hierarchical Clustering, several actionable insights emerge that can guide **OnSports' fantasy player pricing, marketing, and engagement strategies**. These recommendations aim to translate analytical results into measurable business value.

1. Implement Tier-Based Player Pricing

The clustering results consistently divide players into **two distinct performance tiers**:

- **Cluster 1:** Elite or high-impact performers
- **Cluster 0:** Supporting or average contributors

Recommendation:
Adopt a **tier-based pricing model** that scales non-linearly — with top-tier players priced exponentially higher to reflect their outsized fantasy impact. This approach ensures realism, prevents undervaluation of elite performers, and promotes game balance by rewarding users who take calculated budget risks.

2. Introduce Dynamic Pricing Adjustments

Strong correlations between **Minutes Played, Influence, and Total Points** indicate that fantasy performance is sensitive to real-time player form and match participation.

Recommendation:
Incorporate **dynamic price updates** using rolling performance metrics (e.g., points per 90 minutes or recent 5-match form). This keeps

valuations responsive, sustains user engagement, and aligns pricing with current performance trends rather than static season averages.

3. Position-Specific Player Valuation

The analysis confirms significant **positional variance** — midfielders and forwards dominate attacking returns, while defenders and goalkeepers excel in clean sheets and consistency.

Recommendation:
Develop **position-aware pricing and ranking algorithms**.

- Midfielders and forwards → reward attacking output (goals, assists, threat).
- Defenders and goalkeepers → emphasize defensive reliability (clean sheets, goals conceded).
  This ensures fair valuation across all roles, encouraging diverse team compositions among users.

4. Highlight Elite Performers in Marketing Campaigns

Players in **Cluster 1** consistently display high creativity, influence, and bonus points — making them reliable fantasy assets and ideal figures for user engagement.

Recommendation:
Leverage **Cluster 1 players as marketing ambassadors** in app promotions, featured picks, and social media campaigns. Highlight their stats-driven dominance to build credibility and attract competitive fantasy users seeking data-backed insights.

5. Use Clustering for Predictive Scouting and Retention

The clear performance segmentation achieved through both clustering techniques provides a repeatable framework for monitoring upcoming seasons.

Recommendation:
Deploy similar **clustering analysis every season** to track transitions between clusters — identifying emerging players (moving from Cluster 0 → Cluster 1) and underperformers. This can support both internal analytics and user-facing features such as “Breakout Player Predictions” or “Risers & Fallers” reports.

6. Enhance User Experience Through Data Transparency

Users increasingly prefer platforms that offer **analytical insights** to guide decision-making.

Recommendation:
Integrate **data visualization dashboards** within the OnSports app showing cluster-based player tiers, recent form indicators, and fantasy value trends. This not only empowers users but positions OnSports as a **data-driven, premium fantasy platform** strengthening long-term user trust and retention.